University of Minnesota Macro Notes

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Contents

1 Introduction

This document serves as a collection of concepts covered in graduate level macroeconomics. Specifically, the PhD Macroeconomics sequence ECON 8105 (Kehoe), ECON 8106 (Chari), ECON 8107 (Amador), and ECON 8108 (Luttmer) at the University of Minnesota. If you ever encouter these notes and have questions or spot an error, be sure to reach out to me here. Additionally, I speak a lot in the second person because these notes are written in a manner as to be read by myself in the past.

2 ECON 8105 (Kehoe)

2.1 General Equilibrium

There are two main types of economic modeling strategies that we need to talk about in this section. Arrow-Debreu (AD) representations and Sequential Markets (SM) representations. The crux of the idea is to develop a theory for the dynamic interactions of various households (and other sorts of agents) who face various resource constraints and optimization choices. We pose this question as simultaneously solving multiple maximization problems. The AD and SM representations are techniques of translating the structure of economies into a specific mathematical formulation with prices (formed based off of the relative utility preferences of agents).

In the AD representation, households are able to trade in futures contacts for some single consumption good at time zero (before the economy gets up and running) and there is no other specific mechanism to save the consumption good. The also are often able to trade futures contracts for bonds and capital as well depending on the model. We are often asked to describte the structure of such a market and the official wording is below:

Definition 1. *[Arrow Debreu Market Structure] With an Arrow-Debreu markets structure futures, markets for goods are open in period* 0 *. Consumers trade futures contracts among themselves.*

In the SM representation, households are able to trade one-period-ahead futures contracts for a consumption good in each time period.

Definition 2. *[Sequential Markets Structure] With sequential markets market structure, there are markets for goods and bonds open every period. Consumers trade goods and bonds among themselves.*

For the context of this class, one key detail that always tripped me up is that whenever there is capital, there is also a firm (which often goes unspecified) that utilizes this capital and transforms it into the consumption good. In this case, the market structure shifts to households also interacting with the firm who buys claims on capital and labor in exchange for claims on the consumption good. With sequential markets, there are also bonds that households trade amongst themselves.

2.1.1 The Paradox of General Equilibrium

Finally, I wanted to speak a little about the simultaneity of determining all the components of an equilibrium. It is a feature of the economics that is difficult to really intuit and appreciate in the beginning but comes as you work with more models and do more problems. The main confusion is that a lot of times we discuss economics as a series of cause and effects, whereas in these equilibrium models, everything is intertwined. It seems somewhat paradoxical that both households and firms consider prices to be a truth of the world that is independent of their behavior but somehow is also pinned down exactly through the preferences/characteristics of households and firms.

In the math, the simultaneity is clear, we solve all of the maximization problems and try to find such a solution that fit's all the agents in the economy while also satisfying the market clearing conditions. Qualitatively it took me some time to come to terms with how to interpret it until (I am a big fan of sci-fi) I thought about it in the context of the short story "Stories of your life" (it is also the basis for the movie "Arrival" and also a quick read for anyone who is interested) and how it describes the worldview of the aliens through various mathematical problems where there is an analogy between the physical causal interpretation and the simultaneity interpretation through Fermat's Principle of Least Time.

We do a lot of causal analysis in economics, even using general equilibrium models, and I think overcoming this paradox of how the model pins down each of the important objects versus the exercise of causal analysis that we are trying to do is important for getting a foundation of intuition. It will come as you are more involved in it, but I had a easier time with it when I tried to draw analogies to other concepts. The idea is that yes, everything interacts with everything else through some channel or the other, but we can try to wean out to what extent is each channel important.

2.2 Basic Endowment Model

Consider an economy that consists of two households who live forever and want to maximize their lifetime utility. Let $u_i(\cdot)$ be the in-period utility function for agent $i \in \{1,2\}$. Given a consumption stream $\{x_t^i\}_{t=0}^{\infty}$ (where each x_t^i represents the amount of consumption that person i does at time t), we can write the total lifetime utility of agent i as

$$\sum_{t=0}^{\infty} \beta^t u_i(x_t^i).$$

The reason as to why we include this β^t discounting is one of these types of details that does not seem immediately obvious but soon becomes so after working a few problems. In short, it is meant to make sure that infinite sequences (since these agents live forever) converge so that lifetime utility is a well defined object.

Now let $\{w_t^i\}_{t=0}^{\infty}$ be the endowment stream of person i, where each w_t^i represents the amount of the consumption good that person i recieves at time t. Each household i is constrained such that they can only consume as much as their lifetime utility allows. This description of the model may seem a little vague right now, but hopefully a description of the equilibria will pin it down.

Definition 3. *[Arrow Debreu Equilibrium for basic endowment] An Arrow Debreu market equilibrium for such an economy as above is a collection of household allocations* $\{(\hat{c}_t^1, \hat{c}_t^2)\}_{t=0}^{\infty}$ *and prices* $\{\hat{p}_t\}$ *such* *that*

- <u>*Household Maximization:*</u> *Given prices and endowment stream $\{w_t^i\}_{t=0}^{\infty}$, the household allocations solve the household's maximization problem for each i*

$$\begin{aligned}
&\max_{\{c_t^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u_i(c_t^i) \\
s.t. \quad &(1) \quad \sum_{t=0}^{\infty} \hat{p}_t c_t^i \leq \sum_{t=0}^{\infty} \hat{p}_t w_t^i, \\
&(2) \quad c_t^i \geq 0, \forall t \geq 0.
\end{aligned}$$

- <u>*Market Clearing:*</u> *Given endowment streams $\{(w_t^1, w_t^2)\}_{t=0}^{\infty}$, the household allocations also satisfy*

$$(1) \quad \hat{c}_t^1 + \hat{c}_t^2 = w_t^1 + w_t^2.$$

One key difference between the AD and Sequential Markets formulations is that we need to rule out schemes where an individual can keep on borrowing forever and paying but their loans in the current period by borrowing more from future periods (aka a Ponzi scheme). This means we need to also say that there is some borrowing limit that is sufficiently large so that it wouldn't affect any behavior other than disrupting Ponzi schemes.

Definition 4. *[Sequential Markets Equilibrium for basic endowment] A Sequential Markets equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t^1, \hat{b}_{t+1}^1, \hat{c}_t^2, \hat{b}_{t+1}^2)\}_{t=0}^{\infty}$ and prices $\{\hat{r}_t\}$ such that*

- <u>*Household Maximization:*</u> *Given prices, endowment stream $\{w_t^i\}_{t=0}^{\infty}$, and initial bond holdings b_0^i, the household allocations solve the household's maximization problem for each i*

$$\begin{aligned}
&\max_{\{c_t^i, b_{t+1}^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u_i(c_t^i) \\
s.t. \quad &(1) \quad c_t^i + b_{t+1}^i \leq \hat{r}_t b_t^i + w_t^i, \forall t \geq 0, \\
&(2) \quad c_t^i \geq 0, \forall t \geq 0, \\
&(3) \quad b_t^i \geq -B, \textit{ for some sufficiently large } B.
\end{aligned}$$

- <u>*Market Clearing:*</u> *Given endowment streams $\{(w_t^1, w_t^2)\}_{t=0}^{\infty}$, the household allocations also satisfy*

$$\begin{aligned}
&(1) \quad \hat{c}_t^1 + \hat{c}_t^2 = w_t^1 + w_t^2, \\
&(2) \quad \hat{b}_t^1 + \hat{b}_t^2 = 0.
\end{aligned}$$

2.2.1 Equivalence of Arrow-Debreu and Sequential Markets

Both of these markets are indeed equivalent after doing some transformations between the sets of equilibria objects. The main idea is that the allocations will always be the same but the prices, given of course that the prices relay different mechanisms of trading, require some transformation. The exact equivalence will be dependent on the economy structure, but the general idea in deriving the equivalence is to take advantage of the euler equations (I'll point it out clearly below but in general this is the equation that we can back out of the households' first order conditions that intertemporaly relates consumption, aka relates c_t to c_{t+1}) and the fact that the allocations are the same across both sets of equilibria. You can find out a specific worked out example of how this equivalence comes to be in the exercises.

Proposition 1. *[Equivalence of AD and Sequential Markets Equilibrium] Consider the AD equilibrium for the economy above, a collection of household allocations household allocations* $\{(\hat{c}_t^1, \hat{c}_t^2)\}_{t=0}^{\infty}$ *and prices* $\{\hat{p}_t\}$*. With this, we can write that the corresponding sequential markets equilibrium of household allocations* $\{(\bar{c}_t^1, \bar{b}_{t+1}^1, \bar{c}_t^2, \bar{b}_{t+1}^2)\}_{t=0}^{\infty}$ *and prices* $\{\bar{r}_t\}$ *is*

- $\bar{c}_t^i = \hat{c}_t^i, \forall t, i$
- $\bar{r}_t = \frac{\hat{p}_{t-1}}{\hat{p}_t} - 1$
- $\bar{b}_0 = 0$*, and* $\bar{b}_{t+1} = w_t^i + (1 + \bar{r}_t) - \bar{c}_t^i$

We can back out the reverse relationship through the relationships decribed above.

2.3 Basic Optimal Growth Model

Now, consider an economy with a representative household and a representative firm. The household now has the ability to sell their labor and capital to the firm which it uses as inputs to create the consumption good via technology $F(\cdot)$. In return, the firm pays the household wages (for labor) and rent (for capital). It will be a bit more clear once we talk about the equilibrium concepts. For this basic version, the firm acts as if it is operating in a perfectly competitive market which means that it makes zero profit and minimizes costs. Assuming that w_t is the wage rate for labor and r_t^k is the rent for capital, the profits for the firm are $p_t F(k_t, \ell_t) - w_t \ell_t - r_t^k k_t$. We can back out the optimality conditions of the firm from this formulation and will use this in our definition of the equilibrium.

Each household has a constant endowment of labor each period which we will normalize to one. This means that they can spend their time on either working for a wage or something else (the effect of which is captured in the utility function).

One final important detail to hash out is how the capital market works. Households invest in new capital at time t which they can rent out to firms at time $t + 1$. This capital sticks around over time but depreciates with rate δ. It is NOT the case that capital acts like a bond where households lend a certain amount to the firm in time period t and gets paid with interest in time period $t + 1$.

Definition 5. *[Arrow Debreu Equilibrium for basic growth model] An Arrow Debreu market equilibrium for such an economy as above is a collection of household allocations* $\{(\hat{c}_t, \hat{k}_{t+1}, \hat{\ell}_{t+1})\}_{t=0}^{\infty}$*, firm allocations* $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$*, and prices* $\{\hat{p}_t, \hat{r}_t^k, \hat{w}_t\}$ *such that*

- *Household Maximization: Given prices and initial level of capital (bestowed by some higher power)* k_0*, the household allocations solve the household's maximization problem for each i*

$$\begin{aligned} &\max_{\{c_t, \ell_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t) \\ &s.t. \quad (1) \quad \sum_{t=0}^{\infty} \hat{p}_t c_t + \hat{p}_t k_{t+1} - \hat{p}_t (1-\delta) k_t \leq \sum_{t=0}^{\infty} \hat{w}_t \ell_t + \hat{r}_t^k k_t, \\ &\qquad\quad (2) \quad c_t \geq 0, k_t \geq 0, 0 \leq \ell_t \leq 1, \forall t \geq 0. \end{aligned}$$

- *Firm Maximization: Given prices, firms minimize costs and make zero profits, specifically the firm allocations satisfy*

$$\begin{aligned} \frac{\hat{r}_t^k}{\hat{p}_t} &= \frac{\partial F}{\partial k}(\hat{k}_t^f, \hat{\ell}_t^f) \\ \frac{\hat{w}_t}{\hat{p}_t} &= \frac{\partial F}{\partial w}(\hat{k}_t^f, \hat{\ell}_t^f) \end{aligned}$$

- *Market Clearing: The household and firm allocations also satisfy*

$$\begin{aligned} \textit{Goods Market:} &\quad \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1-\delta)\hat{k}_t, \forall t \geq 0, \\ \textit{Capital Market:} &\quad \hat{k}_t = \hat{k}_t^f, \forall t \geq 0, \\ \textit{Labor Market:} &\quad \hat{\ell}_t = \hat{\ell}_t^f, \forall t \geq 0. \end{aligned}$$

Of course, in the sequential markets definition we have bonds. So far, we have described a model with a representative consumer which means that the household has no one to trade bonds with and as a result bond holdings are trivially 0 at any point in time. This will change as we add multiple types of households or some other agent, such as a government, that is willing to buy/sell bonds.

Definition 6. *[Sequential Markets Equilibrium for basic growth model] A sequential markets equilibrium for such an economy as above is a collection of household allocations* $\{(\hat{c}_t, \hat{k}_{t+1}, \hat{b}_{t+1}, \hat{\ell}_{t+1})\}_{t=0}^{\infty}$*, firm allocations* $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$*, and prices* $\{\hat{p}_t, \hat{r}_t^k, \hat{r}_t^b, \hat{w}_t\}$ *such that*

- *Household Maximization: Given prices and initial level of capital (bestowed by some higher power)* k_0 *and bond holdings* b_0*, the household allocations solve the household's maximization problem for* *each i*

$$\max_{\{c_t,\ell_t,k_{t+1},b_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t,\ell_t)$$
$$\begin{aligned} s.t. \quad & (1) \quad c_t + k_{t+1} + \hat{b}_{t+1} \le (1-\delta+\hat{r}_t^k)k_t + \hat{w}_t\ell_t + (1+\hat{r}_t^b)b_t, \\ & (2) \quad c_t \ge 0, k_t \ge 0, 0 \le \ell_t \le 1, \forall t \ge 0, \\ & (3) \quad b_t \ge -B, \textit{ for some sufficiently large } B. \end{aligned}$$

- *Firm Maximization:* *Given prices, firms minimize costs and make zero profits, specifically the firm allocations satisfy*

$$\frac{\hat{r}_t^k}{\hat{p}_t} = \frac{\partial F}{\partial k}(\hat{k}_t^f, \hat{\ell}_t^f)$$
$$\frac{\hat{w}_t}{\hat{p}_t} = \frac{\partial F}{\partial w}(\hat{k}_t^f, \hat{\ell}_t^f)$$

- *Market Clearing:* *The household and firm allocations also satisfy*

$$\begin{aligned} \textit{Goods Market:} \quad & \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1-\delta)\hat{k}_t, \forall t \ge 0, \\ \textit{Capital Market:} \quad & \hat{k}_t = \hat{k}_t^f, \forall t \ge 0, \\ \textit{Labor Market:} \quad & \hat{\ell}_t = \hat{\ell}_t^f, \forall t \ge 0, \\ \textit{Bonds Market:} \quad & \hat{b}_{t+1} = 0, \forall t \ge 0. \end{aligned}$$

2.4 Basic Overlapping Generations Model

An overlapping generations model is not necessarily a separate entity from the concept of the topics we have discussed so far. The idea is still to solve for an equilibrium but the main difference from the models we will discuss in this section versus the section above is that households only live for two periods (something that is often extended in a lot of other models in use right now but the basic version we cover will only have two periods of life) and recieve different endowments in their young and old ages.

Specifically, consider a world with time periods $t \in (1, 2, \ldots)$. In each period, there are two households: one old and one young. The old household will die in the next period and the young household will live for one more period and become old. In this sense, we imbue some heterogeneity between households which often time leads to trading taking place. One important piece of notation is that we will designate an allocation or endowment that belongs to an individual born at time t at time t and $t+1$ respectively as c_t^t, c_{t+1}^t. Or in other words, the subscript denotes the time period in which a certain object belongs to and the superscript denotes the time period in which the person was born. We will also say that in period 1, there exists some person who was born in period 0 that starts out as some old guy in period 1. We don't recognize anything that happened in period 0 or before. This is just to make sure that there are always one old and young household in the economy at any time period $t \geq 1$.

Whether or not trading takes place is an important question in this model. If the old generation has a higher income than the younger generation, then they want to consume the entirety of their income given that they will die in the next period and not be able to recoup any returns from bonds; this often times leads to no trading occuring in the model and there are some exercises later in this section that will demonstrate that.

We can also do growth in an overlapping generations model as well, but I won't cover that in this guide since there aren't many differences between the OLG and infinitely lived versions. As with the previous model, let's look at what an AD and Sequential Markets Equilibrium looks like

Definition 7. *[Arrow Debreu Equilibrium for basic OLG model] An Arrow Debreu Equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t^t, \hat{c}_{t+1}^t)\}_{t=0}^{\infty}$ and prices $\{\hat{p}_t\}_{t=0}^{\infty}$ such that*

- *Initial Old Household Maximization: Given price $\hat{p}_1$ and endowments (m, w_1^0), the household allocation $\hat{c}_1^0$ solves the following maximization problem*

$$\begin{aligned} \max_{c_1^0} \quad & u(c_1^0) \\ s.t. \quad (1) \quad & \hat{p}_1 c_1^0 \leq \hat{p}_1 w_1^0 + m, \\ (2) \quad & c_1^0 \geq 0. \end{aligned}$$

- *Household Maximization: Given prices and endowment structure $\{(w_t^t, w_{t+1}^t\}_{t=0}^{\infty}$, the household allocations solve the series of maximization problems for every $t \geq 1$*

$$\begin{aligned} \max_{c_t^t, c_{t+1}^t} \quad & u(c_t^t, c_{t+1}^t) \\ s.t. \quad (1) \quad & \hat{p}_t c_t^t + \hat{p}_{t+1} c_{t+1}^t \leq \hat{p}_t w_t^t + \hat{p}_{t+1} w_{t+1}^t, \\ (2) \quad & c_t^t, c_{t+1}^t \geq 0. \end{aligned}$$

- *Market Clearing: The allocations also satisfy*

$$\textit{Goods Market:} \quad \hat{c}_t + \hat{c}_{t-1}^t = w_t^t + w_{t-1}^t$$

Then for the sequential markets representation, we add one-period bonds.

Definition 8. *[Sequential Markets Equilibrium for basic OLG model] A Sequential Markets Equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t^t, \hat{c}_{t+1}^t, \hat{b}_{t+1}^t)\}_{t=0}^{\infty}$ and prices $\{\hat{r}_{t+1}\}_{t=0}^{\infty}$ such that*

- *Initial Old Household Maximization: Given price and endowments (m, w_1^0), the household allocation*

$\hat{c}_1^0$ *solves the following maximization problem*

$$\begin{aligned}
\max_{c_1^0} \quad & u(c_1^0) \\
s.t. \quad (1) \quad & c_1^0 \le w_1^0 + m, \\
(2) \quad & c_1^0 \ge 0.
\end{aligned}$$

- *Household Maximization: Given prices and endowment structure $\{(w_t^t, w_{t+1}^t\}_{t=0}^{\infty}$, the household allocations solve the series of maximization problems for every $t \ge 1$*

$$\begin{aligned}
\max_{c_t^t, c_{t+1}^t, b_{t+1}^t} \quad & u(c_t^t, c_{t+1}^t) \\
s.t. \quad (1) \quad & c_t^t + b_{t+1}^t \le w_t^t, \\
(2) \quad & c_t^{t+1} \le w_{t+1}^t + \hat{r}_{t+1} b_{t+1}^t, \\
(3) \quad & c_t^t, c_{t+1}^t \ge 0.
\end{aligned}$$

- *Market Clearing: The allocations also satisfy*

$$\begin{aligned}
\textit{Goods Market:} \quad & \hat{c}_t + \hat{c}_{t-1}^t = w_t^t + w_{t-1}^t \\
\textit{Bonds Market:} \quad & \hat{b}_2^1 = m, \quad \hat{b}_{t+1}^1 = \Pi_{\tau=2}^t (1 + \hat{r}_\tau) m
\end{aligned}$$

2.5 Other Notes

2.5.1 Algebra and Useful Places to Look

It's tough to talk vaguely about algebra, so the goal is to cover at least one of each type of problem you may face in the prelim. I've found that in most cases, for those models that we are able to solve analytically, we can usually do so by backing out the euler condition from the first order conditions and then plugging in this expression into the market clearing conditions. This usually pins down consumption and from this we can look back to the euler condition to determine prices.

The final note of precaution is to be careful about the inada conditions. The inada conditions are a set of conditions what we look at for the utility function of households (but more generally for any sort of maximization problem) that allow us to disregard any lagrangian maximization edge cases. I will cover at least one problem where the inada conditions don't hold and how we work through it.

Definition 9. *[Inada Conditions] Consider any utility function $U(C_t)$. If $\lim_{C_t \to \infty} \frac{\partial U}{\partial C_t} = 0$ and $\lim_{C_t \to 0} \frac{\partial U}{\partial C_t} = \infty$ then the equilibrium value $C_t^* \in (0, +\infty)$.*

2.5.2 Transfers and the Negishi Method

I will come back to this after prelims given that I don't think he typically asks questions about this.

2.5.3 Pareto Optimality Analysis

Pareto optimality is a descriptor used for a specific feasible allocation that is no worse than any other feasible allocation with respect to overall welfare. We can define welfare in any manner of ways but generally in this class we take it to be some weighted sum of the lifetime utility of all households. There are two types of questions we could be faced with

- Solve for optimal allocations as a social planner based on some pareto weights.
- Identify if a particular competitive equilibrium is optimal or not (usually with respect to OLG models given that these are not complete markets). Prove or disprove.

2.5.4 The Time Varying Constraint

We will see why more clearly in the Chari section and conditions that ensure that dynamic programming works, but we also require that asset accumulation, be it capital or bonds (or any other type of asset), doesn't blow up over time relative to prices. The most generalized version of this is the expression

$$\lim_{T\to\infty} \beta^T \lambda_T a_{T+1} = 0.$$

Here λ_T is the lagrange multiplier on the time T budget constraint. The condition itself says that the present discounted value of wealth must go to zero as $T \to \infty$. There are two ways it can fail which have their own implications:

- $a_{T+1} \to +\infty$: the consumer is dying with leftover wealth — they over-saved, which can't be optimal since they could have consumed more.
- $a_{T+1} \to -\infty$: the consumer is running a Ponzi scheme — borrowing without bound, which lenders won't allow (this is ruled out by the no-Ponzi condition separately).

2.6 Exercises

Note for myself: Go back to each section and make clear which questions they should refer to corresponding to each section!

2.6.1 Prelim: Spring 2025

Consider an economy with a representative consumer with the utility function

$$\sum_{t=0}^{\infty} \beta^t \log c_t$$

where $0 < \beta < 1$. This consumer has an endowment of $\bar{\ell}_t = 1$ units of labor in each period and $\bar{k}_0$ units of capital in period 0. Feasible allocation/production plans satisfy

$$c_t + k_{t+1} \leq \theta k_t^{\alpha} \ell_t^{1-\alpha}.$$

(a) Describe an Arrow-Debreu market structure for this economy, explaining when markets are open, who trades with whom, and so on. Define a sequential markets equilibrium.

SOLUTION: An **Arrow Debreu market structure** for this economy is one where futures markets to trade the consumption good, capital, and labor opens in period 0. The representative consumer buys consumption good futures contracts from the representative firm and sell futures contracts for labor and capital to the firm.

An **Arrow Debreu equilibrium** for this economy is a collection of household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f.\hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ such that:

- Household Maximization: Given prices and initial household capital holdings $\bar{k}_0$, the household allocations satisfy the following maximization problem

$$\begin{aligned}
&\max_{c_t,k_t,\ell_t} && \sum_{t=0}^{\infty} \beta^t \log c_t \\
&s.t. \quad (1) && \sum_{t=0}^{\infty} \hat{p}_t c_t + \hat{p}_t k_{t+1} - \hat{p}_t(1-\delta)k_t \leq \sum_{t=0}^{\infty} \hat{r}_t k_t + \hat{w}_t \ell_t, \\
&\qquad\; (2) && c_t, k_t \geq 0, \quad 0 \leq \ell_t \leq 1, \\
&\qquad\; (3) && k_0 = \bar{k}_0.
\end{aligned}$$

- Firm Problem: Given prices, the firm allocations solve the firm's cost minimization problem satisfy the following two conditions

$$\frac{\hat{r}_t}{\hat{p}_t} = F_k(\hat{k}_t^f, \hat{\ell}_t^f),$$

$$\frac{\hat{w}_t}{\hat{p}_t} = F_\ell(\hat{k}_t^f, \hat{\ell}_t^f).$$

- Market Clearing: The allocations also satisfy the following market clearing conditions

$$\begin{aligned}
[Goods] &: \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1-\delta)\hat{k}_t, \\
[Labor] &: \hat{\ell}_t = \hat{\ell}_t^f, \\
[Capital] &: \hat{k}_t = \hat{k}_t^f.
\end{aligned}$$

(b) Describe a sequential markets structure for this economy, explaining when markets are open, who trades with whom, and so on. Define a sequential markets equilibrium.

SOLUTION: A **sequential markets structure** for this economy is one where markets for goods, labor, one-period-ahead bonds, and capital opens every period. Here the representative consumer trivially trades bonds with themselves, buys the consumption good from the firm and sells their labor and capital to the firm.

A **sequential markets equilibrium** for such an economy as above is a collection of household allocations $\{(\hat{c}_t, \hat{k}_{t+1}, \hat{b}_{t+1}, \hat{\ell}_{t+1})\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{r}_t^k, \hat{r}_t^b, \hat{w}_t\}$ such that

- Household Maximization: Given prices and initial level of capital $\bar{k}_0$ and bond holdings $\bar{b}_0$, the household allocations solve the household's maximization problem

$$
\begin{aligned}
\max_{\{c_t, \ell_t, k_{t+1}, b_{t+1}\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t \log c_t \\
s.t. \quad & (1) \quad c_t + k_{t+1} + \hat{b}_{t+1} \leq (1 - \delta + \hat{r}_t^k) k_t + \hat{w}_t \ell_t + (1 + \hat{r}_t^b) b_t, \\
& (2) \quad c_t \geq 0, k_t \geq 0, 0 \leq \ell_t \leq 1, \forall t \geq 0, \\
& (3) \quad b_t \geq -B, \text{ for some sufficiently large } B, \\
& (4) \quad k_0 = \bar{k}_0, b_0 = \bar{b}_0.
\end{aligned}
$$

- Firm Maximization: Given prices, firms minimize costs and make zero profits, specifically the firm allocations satisfy

$$\hat{r}_t^k = F_k(\hat{k}_t^f, \hat{\ell}_t^f)$$

$$\hat{w}_t = F_w(\hat{k}_t^f, \hat{\ell}_t^f)$$

- Market Clearing: The household and firm allocations also satisfy

$$
\begin{aligned}
\text{Goods Market:} \quad & \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1 - \delta) \hat{k}_t, \forall t \geq 0, \\
\text{Capital Market:} \quad & \hat{k}_t = \hat{k}_t^f, \forall t \geq 0, \\
\text{Labor Market:} \quad & \hat{\ell}_t = \hat{\ell}_t^f, \forall t \geq 0, \\
\text{Bonds Market:} \quad & \hat{b}_{t+1} = 0, \forall t \geq 0.
\end{aligned}
$$

■

(c) Carefully state a proposition or propositions that establish the essential equivalence of the equilibrium concept in part a with that in part b. Be sure to specify the relationships between the objects in the Arrow-Debreu equilibrium and those in the sequential markets equilibrium.

SOLUTION: Let's do AD $\rightarrow$ Sequential first. Consider the set of household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ to be an Arrow Debreu equilibrium. We can then write the corresponding Sequential Markets equilibrium of household allocations $\{(\bar{c}_t, \bar{k}_{t+1}, \bar{b}_{t+1}, \bar{\ell}_{t+1})\}_{t=0}^{\infty}$, firm allocations $\{\bar{k}_t^f, \bar{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\bar{r}_t^k, \bar{r}_t^b, \bar{w}_t\}$ to be:

- $\hat{c}_t = \bar{c}_t$, $\hat{k}_t = \bar{k}_t$, $\hat{\ell}_t = \bar{\ell}_t$, $\hat{\ell}_t^f = \bar{\ell}_t^f$, $\hat{k}_t^f = \bar{k}_t^f$ for all $t \geq 0$.
- $\frac{\hat{w}_t}{\hat{p}_t} = \bar{w}_t$, $\frac{\hat{r}_t}{\hat{p}_t} = \bar{r}_t^k$, and $\bar{r}_t^b = \bar{r}_t^k - 1$ for all $t \geq 0$.
- For the economy above, we have that $b_t = 0$ for all $t \geq 0$.

Now, for the opposite direction, consider the Sequential Markets equilibrium of household allocations $\{(\bar{c}_t, \bar{k}_{t+1}, \bar{b}_{t+1}, \bar{\ell}_{t+1})\}_{t=0}^{\infty}$, firm allocations $\{\bar{k}_t^f, \bar{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\bar{r}_t^k, \bar{r}_t^b, \bar{w}_t\}$. We can then write the corresponding Arrow Debreu equilibrium of household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ to be:

- $\hat{c}_t = \bar{c}_t$, $\hat{k}_t = \bar{k}_t$, $\hat{\ell}_t = \bar{\ell}_t$, $\hat{\ell}_t^f = \bar{\ell}_t^f$, $\hat{k}_t^f = \bar{k}_t^f$ for all $t \geq 0$.
- Set numeraire $\hat{p}_0 = 1$, then we also have $\hat{p}_t = \Pi_{i=1}^{t} \frac{1}{\bar{r}_t^k}$
- $\hat{r}_t = \hat{p}_t \bar{r}_t^k$ and $\hat{w}_t = \hat{p}_t \bar{w}_t$

■

(d) Define a Pareto efficient allocation/production plan. Prove either that an Arrow-Debreu allocation/production plan is Pareto efficient or that a sequential markets allocation/production plan is Pareto efficient.

SOLUTION: A pareto efficient allocation/production plan is a set of household alloctions $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$such that it is feasible, meaning that it satisfies the following condition

$$\hat{c}_t + \hat{k}_{t+1} \leq \theta \hat{k}_t^{\alpha} \hat{\ell}_t^{1-\alpha},$$

and also that there exists no other set of feasible household allocations $\{\bar{c}_t, \bar{k}_t, \bar{\ell}_t\}_{t=0}^{\infty}$

where
$$\sum_{t=0}^{\infty} \beta^t \log \hat{c}_t < \sum_{t=0}^{\infty} \beta^t \log \bar{c}_t.$$

Let's now prove that an AD allocation is Pareto Efficient. To do so, let household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ be an AD equilibrium and assume for the sake of contradiction that there exists some other allocation $\{\bar{c}_t, \bar{k}_t, \bar{\ell}_t\}_{t=0}^{\infty}$ that is both feasible and achieves a higher lifetime utility than the allocation from the AD equilibrium.

From the inherent definition of optimization problems, if
$$\sum_{t=0}^{\infty} \beta^t \hat{c}_t < \sum_{t=0}^{\infty} \bar{c}_t$$

then we must have that
$$\sum_{t=0}^{\infty} \hat{p}_t(\bar{c}_t + \hat{k}_{t+1} - (1-\delta)\hat{k}_t) > \sum_{t=0}^{\infty} \hat{r}_t \bar{k}_t + \hat{w}_t \bar{\ell}_t,$$

because if otherwise, then the bar allocations would have solved the optimization problem instead of the hat allocations.

Now, assuming that the firm has the same CRS production as in the feasibility constraint and using the fact that we are assuming competitive markets in which the firm makes 0 profit at equilibrium, we know that
$$\pi_t(\bar{k}_t^f, \bar{\ell}_t^f) = \hat{p}_t \theta (\bar{k}_t^f)^{\alpha} (\bar{\ell}_t^f)^{1-\alpha} - \hat{r}_t \bar{k}_t - \hat{w}_t \bar{\ell}_t \leq 0$$

from the optimality of $\hat{k}_t^f, \hat{\ell}_t^f$. This then let's us write (also assuming that $\delta = 1$)
$$\sum_{t=0}^{\infty} \hat{p}_t(\bar{c}_t + \bar{k}_{t+1}) > \sum_{t=0}^{\infty} \hat{r}_t \bar{k}_t + \hat{w}_t \bar{\ell}_t \geq \sum_{t=0}^{\infty} \hat{p}_t \theta (\bar{k}_t^f)^{\alpha} (\bar{\ell}_t^f)^{1-\alpha} = \sum_{t=0}^{\infty} \hat{p}_t \theta \bar{k}_t^{\alpha} \bar{\ell}_t^{1-\alpha}.$$

This, however, implies that $\bar{c}_t + \bar{k}_{t+1} > \theta \bar{k}_t^{\alpha} \bar{\ell}_t^{1-\alpha}$ which violates the feasibility constraint. We have thus reached a contradiction and indeed it must be that the AD equilibrium is pareto efficient. ■

(e) Write down Bellman's equation that defines the value function for the dynamic programming problem that a Pareto efficient allocation/production plan solves. Guess that the value function has the form $V(k) = a_0 + a_1 \log k$ for some yet-to-be-determined constants a_0 and a_1. Solve for the policy function $k' = g(k)$.

Solution: Note first that because labor ℓ does not enter the household's utility function in any manner, we can immediately claim that they will always use up their entire endowment to produce the consumption good; $\ell_t = 1$ for all $t \geq 0$. Second, since we know that the household's utlity satisfies the inada conditions, we can claim that the feasibility constraint will be binding, i.e. that

$$c_t + k_{t+1} = \theta k_t^\alpha \implies c_t = \theta k_t^\alpha - k_{t+1}.$$

With these facts in hand, we can write down the Bellman equation

$$\boxed{V(k) = \max_{k'} \left\{ \log(\theta k^\alpha - k') + \beta V(k') \right\}.}$$

Next, we guess that $V(k) = a_0 + a_1 \log(k) \implies V'(k) = \frac{a_1}{k}$. With this, we can take the first order condition of the RHS of the expression above in order to find the optimal value of k'.

$$\begin{aligned}
\frac{\partial}{\partial k'} \left[\log(\theta k^\alpha - k') + \beta V(k') \right] &= 0 \\
-\frac{1}{\theta k^\alpha - k'} + \beta \frac{a_1}{k'} &= 0 \\
k' &= \beta a_1 (\theta k^\alpha - k') \\
k' &= \frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta}
\end{aligned}$$

Now we just need to determine what the value of a_1 is to be. To do so, we can plug the optimal k' back into the bellman equation.

$$\begin{aligned}
V(k) &= \max_{k'} \left\{ \log(\theta k^\alpha - k') + \beta V(k') \right\}, \\
a_0 + a_1 \log(k) &= \log\left(\theta k^\alpha - \frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta} \right) + \beta a_0 + \beta a_1 \log\left(\frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta} \right), \\
a_0 + a_1 \log(k) &= \log\left(\frac{\theta k^\alpha}{1 + a_1 \beta} \right) + \beta a_0 + \beta a_1 \log\left(\frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta} \right), \\
a_0 + a_1 \log(k) &= \beta a_0 + \log\left(\frac{\theta}{1 + a_1 \beta} \right) \beta a_1 \log\left(\frac{a_1 \beta \theta}{1 + a_1 \beta} \right) + \alpha \log(k) + \beta a_1 \alpha \log(k), \\
a_1 &= \alpha + \beta \alpha a_1, \\
a_1 &= \frac{\alpha}{1 - \beta \alpha}.
\end{aligned}$$

Now, we can finally classify the policy function as

$$\boxed{k' = g(k) = \alpha \beta \theta k^\alpha.}$$

■

(f) Use the policy function from part e to calculate the sequential markets equilibrium.

SOLUTION: We know from the get go using the bonds market clearing condition that $\bar{b}_t = 0$ for all $t \geq 0$. We also already know from the explanation above regarding labor that $\hat{\ell}^f_t = \hat{\ell}_t = 1$ for all $t \geq 0$.

Next, since we are given $\bar{k}_0$, we can use this in conjunction with the policy function above to first determine the equilibrium capital is

$$\hat{k}^f_t = \hat{k}_t = (\alpha\beta\theta)^{\sum_{s=0}^{t-1}\alpha^s}\,\bar{k}_0^{\alpha^t}.$$

Now we can back out consumption from the goods market clear condition

$$\hat{c}_t = F(\hat{k}_t, 1) + (1-\delta)\hat{k}_t - \hat{k}_{t+1}.$$

Finally, we have to now compute prices, but these come quite directly from the firm's problem

$$1 + \hat{r}^b_t = \hat{r}^k_t = F_k(\hat{k}_t, 1),$$
$$\hat{w}_t = F_w(\hat{k}_t, 1).$$

■

2.6.2 Prelim: Spring 2023 (Infinitely lived consumers with leisure and dynamic programming)

Consider an economy in which the representative consumer lives forever. There is a good in each period that can be consumed or saved as capital as well as labor. The consumer's utility function is

$$\sum_{t=0}^{\infty} \beta^t u(c_t, x_t).$$

The consumer is endowed with 1 unit of labor in each period, some of which can be consumed as leisure, x_t, and some of which is supplied as labor, ℓ_t. The consumer is also endowed with $\bar{k}_0$ units of capital in period 0. Feasible allocations satisfy

$$c_t + k_{t+1} - (1-\delta)k_t \leq f(k_t, \ell_t).$$

(a) Formulate the problem of maximizing the representative consumer's utility subject to feasibility conditions as a dynamic programming problem. Write down the appropriate Bellman's equation. Make appropriate assumptions on the parameters β and δ and on the functions u and f that guarantee a solution to Bellman's equation. Cite any results that you need, but do not prove anything.

SOLUTION: ∎

(b) Define a sequential markets equilibrium for this economy. Suppose that you have solved the dynamic programming problem in part (a). Explain carefully how to calculate the sequential markets equilibrium.

SOLUTION: ∎

(c) Define an Arrow–Debreu equilibrium for this economy. Suppose that you have solved the dynamic programming problem in part (a). Explain carefully how to calculate the Arrow–Debreu equilibrium.

SOLUTION: ∎

(d) Suppose now that there are equal amounts of two types of consumers in the economy. The two types of consumers have the same discount factor β. They have different utility functions in each period, $u_1(c,x)$ and $u_2(c,x)$, different endowments of labor in each period, $\bar{\ell}^1$ and $\bar{\ell}^2$, and different initial endowments of capital, $\bar{k}_0^1$ and $\bar{k}_0^2$. Define a sequential markets equilibrium.

SOLUTION: ∎

(e) Does the equilibrium allocation for the economy in part (d) solve a dynamic programming problem like that in part (a)? Carefully explain why or why not. If it does solve such a problem, write down the appropriate Bellman's equation.

SOLUTION: ∎

2.6.3 Prelim: Fall 2024 (Pareto efficiency in an overlapping generations economy)

Consider an overlapping generations economy in which there is one good in each period and each generation, except the initial one, lives for two periods. The representative consumer in generation t, $t = 1, 2, \ldots$, has the utility function

$$\log c_t^t + c_{t+1}^t$$

and the endowment $(w_t^t, w_{t+1}^t) = (2, 2)$. [Notice that the utility function is not $\log c_t^t + \log c_{t+1}^t$.] The representative consumer in generation 0 lives only in period 1, has utility function $\log c_1^0$, and has the endowment $w_1^0 = 2$. There is no fiat money.

(a) Define an Arrow–Debreu equilibrium for this economy. Calculate the unique Arrow–Debreu equilibrium.

SOLUTION: ■

(b) Define a sequential markets equilibrium for this economy. Calculate the unique sequential markets equilibrium.

SOLUTION: ■

(c) Define a Pareto efficient allocation. Either prove that the equilibrium allocation in part (a) is Pareto efficient or prove that it is not.

SOLUTION: ■

(d) Suppose now that there is a storage technology available in every period t, $t = 1, 2, \ldots$, that transforms one unit of the good in period t into $\gamma > 0$ units of the good in period $t + 1$. Define a sequential markets equilibrium for this economy. Under what conditions on γ will the storage technology be used in equilibrium?

SOLUTION: ■

3 ECON 8106 (Chari)

3.1 Dynamic Programming (Deterministic and Stochastic)

3.2 Recursive Competitive Equilibria

This is perhaps the concept that I have had the most trouble with in this class. Not because it encapsulates a particularly difficult concept, but more so because it is a lot to remember without the right intuition.

3.3 Cash-Credit-Good Models

3.4 Sticky Prices

3.5 Exercises

There is a very specific set of questions that Chari tends to pull from for the prelim and final/midterm. Here's a quick list

- Cash-Credit Goods Models (with and without Production)
- Recursive Competitive Equilibria
- Search and Unemployment Models

- Sticky Prices
- Island Economy Models
- Sustainable Commitment

3.5.1 Prelim: Spring 2025 (Inflation and Welfare)

Consider a stochastic cash credit goods economy. in which households have preferences over deterministic sequences of the form

$$\sum_{t=0}^{\infty} \beta^t \log c_{1t} + \alpha \log c_{2t}$$

where c_{1t} and c_{2t} denote consumption of cash and credit goods respectively, and $0 < \beta < 1$ is the discount factor. Households are endowed with y units of a composite good which can be converted into cash and credit goods at a 1-1 rate. The resource constraint for deterministic sequences is given by

$$c_{1t} + c_{2t} = y.$$

The endowment y follows a first order Markov process and is the only source of uncertainty in the economy. The securities market meets at the beginning of the period. The household's securities market constraint (for a deterministic version of the economy) is

$$M_t + B_t = (M_{t-1} - p_{t-1}c_{1t-1}) - p_{t-1}c_{2t-1} + p_{t-1}y + R_{t-1}B_{t-1} + T_{t-1}$$

where M_t denotes cash balances, B_t denotes holdings of one-period debt, p_t denotes the price level, R_t denotes the (gross) interest rate on debt and T_t denotes lump-sum transfers by the government. The cash in advance constraint (for a deterministic version of the economy) is

$$p_t c_{1t} \leq M_t$$

Assume real debt holdings are bounded below by a large negative number.

(a) Define a competitive equilibrium. In particular, be precise about what allocations depend on.

SOLUTION: A competitive equilibrium in this economy (assuming complete markets) is a collection of household allocations $\{\hat{c}_{1t}(s^t), \hat{c}_{2t}(s^t), \hat{M}_t(s^t), \hat{B}_t(s^t)\}_{t,s^t}$, government allocations $\{\hat{M}_t^g(s^t), \hat{B}_t^g(s^t), \hat{T}_t(s^t)\}_{t,s^t}$, and prices $\{\hat{p}_t(s^t), \hat{R}_t(s^t)\}_{t,s^t}$ such that:

- Household Maximization: Given prices, endowment y, initial money holdings $\bar{M}_0(s^0)$, initial bond holdings $\bar{B}_0(s^0)$, government transfers, and probabilities of state the household allocations solve the maximization problem

$$\max_{c_{1t}(s^t),c_{2t}(s^t),M_t(s^t),B_t(s^t)} \sum_{t=0}^{\infty}\sum_{s^t}\beta^t(\log c_{1t}(s^t)+\alpha\log c_{2t}(s^t))$$

$$\begin{aligned}
s.t.\quad (1)\quad & M_{t+1}(s^{t+1})+B_{t+1}(s^{t+1})=(M_t(s^t)-\hat{p}_t(s^t)c_{1t}(s^t)) \\
& -\hat{p}_t(s^t)c_{2t}(s^t)+\hat{p}_t(s^t)y_t(s^t) \\
& +\hat{R}_t(s^t)B_t(s^t)+\hat{T}_t(s^t), \\
(2)\quad & \hat{p}_t(s^t)c_{1t}(s^t)\le M_t(s^t), \\
(3)\quad & c_{1t}(s^t),c_{2t}(s^t),M_t(s^t)\ge 0, \\
(4)\quad & B_t(s^t)\ge -B.
\end{aligned}$$

- Government Budget Constraint: Given prices, the government allocations satisfy the following budget clearing constraint for all $t\ge 1$ and s^t

$$\hat{M}_t^g(s^t)+B_t^g(s^t)=\hat{M}_{t-1}^g(s^t)+\hat{R}_{t-1}(s^t)B_{t-1}^g(s^t)+\hat{T}_{t-1}(s^t).$$

- Market Clearing: The allocations also satisfy the following market clearing conditions for all $t\ge 0$ and s^t

$$\begin{aligned}
[Goods]:\quad & \hat{c}_{1t}(s^t)+\hat{c}_{2t}(s^t)=y_t(s^t), \\
[Bonds]:\quad & \hat{B}_t^g(s^t)=\hat{B}_t(s^t), \\
[Money]:\quad & \hat{M}_t^g(s^t)=\hat{M}_t(s^t).
\end{aligned}$$

■

(b) Assume government policy is characterized by a sequence of constant interest rates, $R_t = R > 1$ for all t and for all realizations of the exogenous uncertainty. Characterize the stationary competitive equilibrium.

Solution: Let's first take a look at the household's optimization problem and corresponding first order conditions, given that the utility function satisfies the inada conditions, we can write the lagrangian as

$$\begin{aligned}
\mathcal{L}(c_{1t}(s^t),c_{2t}(s^t),M_t(s^t),B_t(s^t),\lambda_t(s^t),\mu_t(s^t)) = & \sum_{t=0}^{\infty}\beta^t(\log c_{1t}(s^t)+\alpha\log c_{2t}(s^t)) \\
& +\sum_{t=0}^{\infty}\lambda_t(s^t)(\text{long BC})
\end{aligned}$$

$$+ \sum_{t=0}^{\infty} \mu_t(s^t)(M_t(s^t) - p_t(s^t)c_{1t}(s^t)).$$

The corresponding FOCs are then:

$$\begin{aligned}
[c_{1t}(s^t)] &: \frac{\beta^t}{c_{1t}(s^t)} - \lambda_t(s^t)\hat{p}_t(s^t) - \mu_t(s^t)\hat{p}_t(s^t) = 0, \\
[c_{2t}(s^t)] &: \frac{\alpha\beta^t}{c_{2t}(s^t)} - \lambda_t(s^t)\hat{p}_t(s^t) = 0, \\
[M_{t+1}(s_{t+1}, s^t)] &: -\lambda_t(s^t) + \lambda_{t+1}(s_{t+1}, s^t) + \mu_{t+1}(s_{t+1}, s^t) = 0, \\
[B_{t+1}(s_{t+1}, s^t)] &: -\lambda_t(s^t) + R_{t+1}(s_{t+1}, s^t)\lambda_{t+1}(s_{t+1}, s^t) = 0.
\end{aligned}$$

If the cash-in-advance constraint is not binding, then we know that $\mu_{t+1}(s_{t+1}, s^t) = 0$ for all $t \geq 0$ and (s_{t+1}, s^t). From the third FOC, this gives us that $\lambda_t(s^t) = \lambda_{t+1}(s_{t+1}, s^t)$ which further implies (from the fourth FOC) that $R_{t+1}(s_{t+1}, s^t) = R = 1$. This, however, is a contradiction with the government policy that $R > 1$.

Now let's handle the case in which the money in advance constraint is indeed binding, specifically that $M_t(s^t) = p_t(s^t)c_{1t}(s^t)$. Using this, we can rewrite the first FOC as

$$\beta^t - \lambda_t(s^t)M_t(s^t) - \mu_t(s^t)M_t(s^t) = 0 \implies \lambda_t(s^t) + \mu_t(s^t) = \frac{\beta^t}{M_t(s^t)}.$$

Using the third FOC, we then get that

$$\begin{aligned}
-\lambda_t(s^t) + \lambda_{t+1}(s_{t+1}, s^t) + \mu_{t+1}(s_{t+1}, s^t) = 0 &\implies \lambda_t(s^t) = \frac{\beta^{t+1}}{M_{t+1}(s_{t+1}, s^t)}, \\
&\implies \beta^t = \lambda_{t-1}(s^{t-1})p_t(s_t, s^{t-1})c_{1t}(s_t, s^{t-1}).
\end{aligned}$$

Next from the second FOC, we get

$$\beta^t = \frac{\lambda_t(s^t)p_t(s^t)c_{2t}(s^t)}{\alpha}.$$

Dividing these two expressions we get that

$$1 = \frac{\alpha\lambda_{t-1}(s^{t-1})c_{1t}(s_t, s^{t-1}}{\lambda_t(s_t, s^{t-1})c_{2t}(s_t, s^{t-1})} \implies c_{2t}(s^t) = \alpha R c_{1t}(s^t).$$

To get consumption, we can finally plug this into the market clearing for goods.

$$c_{1t}(s^t)\,(1 + \alpha R) = y_t(s^t) \implies \boxed{c_{1t}(s^t) = \frac{1}{1 + \alpha R} y_t(s^t),\ c_{2t}(s^t) = \frac{\alpha R}{1 + \alpha R} y_t(s^t).}$$

Next, to pin down prices, we know from the last FOC that

$$
\begin{aligned}
R &= \frac{\lambda_t(s^t)}{\lambda_{t+1}(s_{t+1}, s^t)}, \\
R &= \frac{\alpha\beta^t}{p_t(s^t)c_{2t}(s^t)} \frac{p_{t+1}(s_{t+1}, s^t)c_{2t+1}(s_{t+1}, s^t)}{\alpha\beta^{t+1}}, \\
\beta R &= \frac{p_{t+1}(s_{t+1}, s^t)y_{t+1}(s_{t+1}, s^t)}{p_t(s^t)y_t(s^t)}, \\
p_t(s_t, s^{t-1}) &= \beta R p_{t-1}(s^{t-1}) \frac{y_{t-1}(s^{t-1})}{y_t(s_t, s^{t-1})}.
\end{aligned}
$$

Now finally, setting $p_0(s_0) = 1$ as the numeraire, we get that prices are

$$
\boxed{p_t(s^t) = p_t(s_t, s^{t-1}) = (\beta R)^t \prod_{s=0}^{\tau-1} \frac{y_\tau(s^\tau)}{y_{\tau+1}(s_{\tau+1}, s^\tau)}.}
$$

Money holdings follows very clearly

$$
\boxed{M_t^g(s^t) = M_t(s^t) = p_t(s^t)c_{1t}(s^t).}
$$

Finally, we can't really pin down $B_t(s^t)$ and $T_t(s^t)$ (we don't have enough equations to do so) other than noting that $T_t(s^t) = B_{t+1}(s_{t+1}, s^t) - RB_t(s^t)$ and $B_t^g(s^t) = B_t(s^t)$.

Keep in mind, everything I have done above assumes complete markets for bonds and money holdings! ■

(c) Compare stationary equilibrium outcomes and welfare for two economies with different constant interest rates, but are otherwise identical.

Solution: Welfare in this economy is simply the sum

$$
\begin{aligned}
&\sum_{t=0}^{\infty} \sum_{s^t} \beta^t (\log c_{1t}(s^t) + \alpha \log c_{2t}(s^t)) \\
&\quad = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \left(\log \left(\frac{y_t(s^t)}{1+\alpha R} \right) + \alpha \log c_{2t} \left(\frac{\alpha R y_t(s^t)}{1+\alpha R} \right) \right) \\
&\quad = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \left[\alpha \log(\alpha R) - (1+\alpha)\log(1+\alpha R) + (1+\alpha)\log(y_s(s^t)) \right].
\end{aligned}
$$

Since both $\alpha < 1 + \alpha$ and $\log(\alpha R) < \log(1 + \alpha R)$ we can say that increases in R decreases welfare. For prices, we can see clearly than an increase in R also increases prices and similarly for money holdings. ■

3.5.2 Prelim: Fall 2025 (A Cash-Credit Goods Model with Private Money)

Consider a cash-credit goods model in which households have preferences of the form $\sum_{t=0}^{\infty} \beta^t (\log c_{1t} + \phi \log c_{2t})$, where c_{1t} and c_{2t} denote consumption of cash and credit goods respectively, ϕ is a parameter, and $0 < \beta < 1$ is the discount factor. Households supply labor inelastically. The resource constraint is given by

$$c_{1t} + c_{2t} + k_{t+1} = k_t^{\alpha} + (1 - \delta) k_t$$

where k_t denotes the capital stock, the capital share parameter α and the depreciation rate δ are both positive and less than 1.

It is convenient to let q_t denote the price of money (or the inverse of the price level). The representative household's budget constraint is given by

$$q_t M_{t+1} + c_{1t} + c_{2t} + k_{t+1} \leq q_t M_t + R_{kt} k_t + w_t l_t + q_t T_t$$

where M_t denotes cash balances, R_{kt} denotes the return to capital and T_t denotes lump sum transfers of cash by the government. The household can use fraction θ of its capital stock to purchase cash goods. The cash in advance constraint is

$$c_{1t} \leq q_t M_t + \theta k_t.$$

(a) Define a competitive equilibrium.

SOLUTION: ■

(b) Assume that the aggregate stock of money grows at a constant rate, π. Characterize the steady state.

SOLUTION: ■

(c) Show that if π is above a critical threshold, the economy has a nonmonetary steady state in which the price of money is zero.

SOLUTION: ■

(d) Show that below the critical threshold, comparing steady states, per capita output is higher if the growth rate of money is higher.

SOLUTION: ■

(e) Do your results in part (d) imply that the Friedman Rule is not optimal in this economy?

SOLUTION: ■

3.5.3 Prelim: Spring 2024 (Search and Unemployment)

Consider the behavior of a worker who seeks to maximize the expected present discounted value of utility given by

$$\sum \beta^t u(c_t).$$

The discount factor is $\beta \in (0, 1)$. In this economy, jobs are destroyed at rate $1 - \theta$, and the worker becomes unemployed. If an individual is unemployed he/she receives $b > 0$ units of consumption (unemployment compensation) if he/she has been unemployed for at most $n - 1$ periods. From the nth period on, an unemployed worker receives $b = 0$. Suppose first that the worker cannot borrow or lend. The worker receives wage draws from a fixed distribution $F(w)$.

A. Consider the situation of an unemployed worker who is in his/her nth period of unemployment (that is, he/she is not collecting unemployment benefits). Argue that the optimal strategy is of the reservation wage variety, and display a formula for the reservation wage. Denote this reservation wage by $w^*(0)$.

SOLUTION: We can write out a bellman equation to describe this unemployed worker's optimization problem,

$$V_{unemp}(w, 0) = \max \left\{ u(0) + \beta \mathbb{E}\left[V_{unemp}(w', 0)\right], V_{emp}(w) \right\}.$$

Here $V_{emp}(w)$ represents the value of worker who is employed at wage w, which we can represent as

$$V_{emp}(w) = u(w) + \beta \left[\theta V_{emp}(w) + (1-\theta)\mathbb{E}\left[V_{unemp}(w', n)\right]\right],$$
$$\implies V_{emp}(w) = \frac{1}{1-\beta\theta}\left[u(w) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)]\right].$$

In $V_{unemp}(w, 0)$, we know that the left hand term in the expression is a constant and the right hand side is increasing w (and also contains the LHS in the range of the RHS for proper distributions $F(\cdot)$). As such, by intermediate value theorem, we know there must exist some $w^*(0)$ such that

$$\beta \mathbb{E}\left[V_{unemp}(w', 0)\right] = V_{emp}(w^*(0)),$$

such that if $w > w^*(0)$ then the worker accepts the wage offer and if $w < w^*(0)$ then the worker declines the offer. To more explicitly define $w^*(0)$, we can write

$$\frac{1}{1-\beta\theta}\left[u(w^*(0))+\beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right] = u(0)+\beta\mathbb{E}\left[V_{unemp}(w',0)\right],$$
$$\implies u(w^*(0))+\beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)] = (1-\beta\theta)\left[u(0)+\beta\mathbb{E}\left[V_{unemp}(w',0)\right]\right].$$

This gives us the expression

$$\boxed{w^*(0) = u^{-1}\left((1-\beta\theta)\left[u(0)+\beta\mathbb{E}\left[V_{unemp}(w',0)\right]-\beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right]\right).}$$

■

B. Consider next the situation of an unemployed worker who is in his/her $(n-1)$th period of unemployment (that is, this is the last period that the worker will collect unemployment compensation). Let the reservation wage of such a worker be $w^*(1)$. Show that $w^*(1) > w^*(0)$ (if you cannot prove it, a serious argument will receive partial credit).

SOLUTION: We can once again write out the bellman equation that represents the problem that this worker faces,

$$V_{unemp}(w,1) = \max\left\{u(b)+\beta\mathbb{E}[V_{unemp}(w',0), V_{emp}(w)]\right\},$$

where $V_{emp}(w)$ is as written above. Once again, we can write

$$u(b)+\beta\mathbb{E}[V_{unemp}(w',0) = V_{emp}(w^*(1)),$$
$$u(b)+\beta\mathbb{E}\left[V_{unemp}(w',0)\right] = \frac{1}{1-\beta\theta}\left[u(w^*(1))+\beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right],$$
$$u(w^*(1))+\beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)] = (1-\beta\theta)\left[u(b)+\beta\mathbb{E}\left[V_{unemp}(w',0)\right]\right].$$

This finally gives us

$$w^*(1) = u^{-1}\left((1-\beta\theta)\left[u(b)+\beta\mathbb{E}\left[V_{unemp}(w',0)\right]\right]-\beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right).$$

Since $b>0$, we know that $u(b)>u(0)$; since we know $u(\cdot)$ is strictly increasing, we know that $u^{-1}(\cdot)$ is also strictly increasing and therefore

$$w^*(1) > w^*(0).$$

■

C. What does the result from (b) say about the probability of leaving unemployment as a function of the duration of the unemployment spell?

SOLUTION: It suggests that the probability of **leaving unemployment increases as the duration of the unemployment spell increases**. We will show later that $w^*(T) > w^*(T-1)$ for all $T < n$. We know that the strategy of the worker boils down to accepting any wage below the reservation wage. Since we know the cdf $F(\cdot)$, we can claim that the probability than an unemployment spell ends is the probability $1 - F(w^*)$. As such

$$\begin{aligned}
&w^*(T) > w^*(T-1), \\
\implies\ &F(w^*(T)) > F(w^*(T-1)), \\
\implies\ &1 - F(w^*(T)) < 1 - F(w^*(T-1)).
\end{aligned}$$

■

D. Let $w^*(2)$ be the reservation wage of a worker who can collect unemployment compensation for two periods before his/her eligibility expires. Show that $w^*(2) > w^*(1)$.

SOLUTION: We can follow a similar train of logic as we did in previous parts. First let's calculate $w^*(2)$ and then go from there

$$\begin{aligned}
u(b) + \beta \mathbb{E}[V_{unemp}(w', 1)] &= V_{emp}(w^*(2)), \\
\frac{1}{1-\beta\theta}\left[u(w^*(2)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)]\right] &= u(b) + \beta \mathbb{E}\left[V_{unemp}(w', 1)\right], \\
(1-\beta\theta)\left[u(b) + \beta \mathbb{E}\left[V_{unemp}(w', 1)\right]\right] &= u(w^*(2)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)].
\end{aligned}$$

From there we can then write

$$w^*(2) = u^{-1}\left((1-\beta\theta)\left[u(b) + \beta \mathbb{E}\left[V_{unemp}(w', 1)\right]\right] - \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)]\right).$$

We can clearly see the similarities in the formulation between $w^*(2)$ and $w^*(1)$ and as such, proving $w^*(2) > w^*(1)$ boils down (after doing some algebra) to proving $\mathbb{E}[V_{unemp}(w', 1)] > \mathbb{E}[V_{unemp}(w', 0)]$. Furthermore we know that $V_{unemp}(w, 1) > V_{unemp}(w, 0)$ for any w would imply $\mathbb{E}[V_{unemp}(w', 1)] > \mathbb{E}[V_{unemp}(w', 0)]$. As such, pick any arbitrary w in the domain of possible w. There are then four cases

- Case 1: If $V_{unemp}(w, 1) = V_{emp}(w)$ and $V_{unemp}(w, 0) = V_{emp}(w)$ then $V_{unemp}(w, 1) = V_{unemp}(w, 0)$
- Case 2: If $V_{unemp}(w, 1) = V_{emp}(w)$ and $V_{unemp}(w, 0) = u(0) + \beta \mathbb{E}[V_{unemp}(w', 0)]$

then

$$V_{unemp}(w,1) = V_{emp}(w) \geq u(b) + \beta \mathbb{E}[V_{unemp}(w',0)] > u(0) + \beta \mathbb{E}[V_{unemp}(w',0)].$$

- Case 3: If $V_{unemp}(w,1) = u(b) + \beta \mathbb{E}[V_{unemp}(w',0)]$ and $V_{unemp}(w,0) = \beta \mathbb{E}[V_{unemp}(w',0)]$ then very simply

$$V_{unemp}(w,1) = u(b) + \beta \mathbb{E}[V_{unemp}(w',0)] > u(0) + \beta \mathbb{E}[V_{unemp}(w',0)].$$

- Case 4: If $V_{unemp}(w,1) = u(b) + \beta \mathbb{E}[V_{unemp}(w',0)]$ and $V_{unemp}(w,0) = V_{emp}(w)$ then again very simply

$$V_{unemp}(w,1) = u(b) + \beta \mathbb{E}[V_{unemp}(w',0)] > V_{emp}(w) = V_{unemp}(w,0).$$

For proper distributions $F(\cdot)$, we know that the measure of w's that lie in cases 2,3, and 4 are non-zero and as such we can conclude that

$$V_{unemp}(w,1) > V_{unemp}(w,0) \implies w^*(2) > w^*(1).$$

■

E. Use an induction argument to show that $w^*(T) > w^*(T-1)$ for all $T \leq n$.

SOLUTION: Let's do a proof by induction

- Base Case: Let $n = 1$, we have shown above that $w^*(1) > w^*(0)$.
- Induction Hypothesis: Assume that for any $k < n$ we have that $w^*(k) > w^*(k-1)$ we want to show that $w^*(k+1) > w^*(k)$.
- Induction Step: Pick any arbitrary $k < n$. As evidenced in the previous problem, we know that

$$w^*(k+1) = (1-\beta\theta)\left[u(b) + \beta \mathbb{E}\left[V_{unemp}(w',k)\right]\right] - \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)].$$

As such, proving $w^*(k+1) > w^*(k)$ is equivalent to proving that $\mathbb{E}\left[V_{unemp}(w',k)\right] > \mathbb{E}\left[V_{unemp}(w',k-1)\right]$. Fix any arbitrary w in the domain. Again, there are three cases

  - Case 1: If $V_{unemp}(w,k) = V_{emp}(w)$ and $V_{unemp}(w,k-1) = V_{emp}(w)$ then $V_{unemp}(w,k) = V_{unemp}(w,k-1)$

– Case 2: If $V_{unemp}(w,k) = V_{emp}(w)$ and $V_{unemp}(w,k-1) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-2)]$ then

$$V_{unemp}(w,k) = V_{emp}(w) \geq u(b) + \beta\mathbb{E}[V_{unemp}(w',k-2)] > V_{unemp}(w,k-1).$$

– Case 3: If $V_{unemp}(w,k) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-1)]$ and $V_{unemp}(w,k-1) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-2)]$. From the induction hypothesis we know that $w^*(k) > w^*(k-1)$ which also implies

$$\mathbb{E}[V_{unemp}(w',k-1)] > \mathbb{E}[V_{unemp}(w',k-2)].$$

This concludes this case.

– Case 4: If $V_{unemp}(w,k) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-1)]$ and $V_{unemp}(w,k-1) = V_{emp}(w)$ then again very simply

$$V_{unemp}(w,k) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-1)] > V_{emp}(w) = V_{unemp}(w,k-1).$$

For proper distributions $F(\cdot)$, we know that the measure of w's that lie in cases 2,3, and 4 are non-zero and as such we can conclude that

$$V_{unemp}(w,k) > V_{unemp}(w,k-1) \implies w^*(k+1) > w^*(k).$$

Thus concludes the proof. ■

F. Suppose now that the worker can save (but not borrow) at an exogenous interest rate R. Set up the dynamic programming problem of the worker. Assume the worker starts with A_0 units of assets and is unemployed. Show (for extra credit) that a worker with larger assets has a higher reservation wage.

Solution: The new dynamic programming problem of an unemployed worker, who has $1 \leq k \leq n$ many periods of unemployment benefits left, a many assets is now, and faces a wage draw w is

$$V_{unemp}(w,a,k) = \max_{0 \leq a' \leq b+a} \left\{ u(b+a-a')) + \beta\mathbb{E}[V_{unemp}(w',Ra',k-1)], V_{emp}(w,a) \right\},$$

with

$$V_{unemp}(w,a,0) = \max_{0 \leq a' \leq a} \left\{ u(a-a') + \beta\mathbb{E}[V_{unemp}(w',Ra',k-1)], V_{emp}(w,a) \right\},$$

and

$$V_{emp}(w,a) = \max_{0 \le a' \le a+w} \left\{ u(w+a-a') + \beta \left[\theta V_{emp}(w, Ra') + (1-\theta)\mathbb{E}[V_{unemp}(w', Ra', n)] \right] \right\}.$$

■

3.5.4 Prelim: Fall 2024 (Mobility, Unemployment, and Equilibrium)

Consider an infinite-horizon economy with a continuum of islands. At the beginning of each period, each island receives a shock z which takes on a finite set of values. The shock is identically, independently distributed across islands, and follows a Markov process with transition probabilities $\pi(z', z)$. In each island, competitive firms operate a technology. Let $f(x,z)$ denote the marginal product of labor on an island with workforce x and current shock z. Assume that f is strictly decreasing in x and strictly increasing in z, and that $f(0,z)$ is strictly positive. Also assume that there is some $\bar{x}$ such that $f(\bar{x}, z) = 0$ for all z. Workers are risk neutral and have preferences given by

$$\sum_{t=0}^{\infty} \beta^t c_t.$$

At the beginning of each period, after observing the realization of z on all the islands, workers in an island indexed by, say, x, z, can choose either to continue working in that island, receiving a wage of $f(x,z)$, or move. If they choose to move, they receive a wage of 0 in the current period, and can move to any island of their choice at the beginning of the next period, after observing the shocks z' in all islands.

Your main task is to define a stationary competitive equilibrium. To do so, conjecture that there are three types of islands in this economy.

A. One type of island has a positive flow of immigrants into it in the current period. Also all current workers stay. The wage must be the same in all these islands. Why? Let θ denote this wage.

B. In the second type of island, all current workers stay and no workers arrive. Why might this happen?

C. The third type of island has some or all current workers leave and no immigrants arrive.

With this conjecture, what is the Bellman equation of workers in the three types of islands? Write a common Bellman equation for all workers.

SOLUTION: Let's carefully go through each of the cases

(IA) In this situation, the wage in the island (x,z) is high enough such that it is higher than the value of leaving and seaching. Crirically the wage θ must be equal in all such islands because if there is some island with wage θ' then no one would choose to go to any island with wage θ and these islands would be of type B instead of type A.

(IB) In this situation, the wage in the island (x, z) is high enough such that it is higher than the value of leaving and seaching but not high enough such people in other islands would do the same. The wage in this island is high but not high enough that others would forgo a period of wages in their own island to come over.

(IC) In this situation the wage in island (x, z) is low enough such that the value of leaving the island, forgoing one time period of wage, and searching for another island is higher than staying in the island and accepting the wage. In theory, it is possible for the wage to be lower than the search value but we will note that in equilibrium it must be that the wage is exaclty the value of searching (which is something that comes from the fact that $f(x, z)$ is strictly decreasing as x increases and visa-versa).

Since islands of Type A all have the same wage, we know that the value of searching for such an island must be equal for a household in any state (x, z). Also note that the wage in a given island must equal the marginal product of labor on that same island. Given θ, the corresponding bellman equation that all workers solve is

$$V(x, z) = \max \left\{ f(x, z) + \beta \sum_{z'} \pi(z', z) V(x', z'), \beta V(x_\theta, z_\theta) \right\}$$

where x_θ, z_θ are any state such that $\theta = f(x_\theta, z_\theta)$.

■

Define a stationary competitive equilibrium. The key object here is a distribution. A distribution over what objects?

SOLUTION: ■

Sketch how you might prove that such an equilibrium exists.

SOLUTION: ■

3.5.5 Prelim: Fall 2023 (The New Classical Phillips Curve)

Consider an infinite-horizon economy cash in advance economy. In each period the economy is affected by a shock s which takes on a finite set of values, and is i.i.d. over time. Consumers have preferences over the single final consumption good, c_t, in each period given by

$$E \sum_{t=0}^{\infty} \beta^t \log c_t$$

and supply one unit of labor inelastically. The final consumption good is an aggregate of intermediate goods c_{it} given by

$$c_t = \left[\int_0^1 c_{it}^{\theta;1/\theta} \right]. \tag{1}$$

The technology for each intermediate good is given by

$$c_{it} = l_{it}, \tag{2}$$

where l_{it} is the amount of labor allocated to intermediate good i in period t. Competitive retail firms use (1) to produce the final good using as inputs the intermediate goods. Intermediate goods firms are monopolists of their particular good. A fraction α of intermediate goods is produced by *sticky price* firms which must set prices for their good in period t before the current shock s_t is realized. These firms are required to meet all the demand at their posted prices. These prices are sticky only for one period at a time. The remaining $1 - \alpha$ of intermediate goods firms set prices after the shock is realized.

Consumers face a cash in advance constraint of the form

$$P_t c_t \leq M_t + T_t \tag{3}$$

where M_t denotes the amount of money brought into period t and T_t denotes lump sum transfers by the government. Consumers have budget constraints of the form

$$P_t c_t + M_{t+1} \leq W_t l_t + M_t + T_t + \Pi_t$$

where P_t is the nominal price of the final consumption good, W_t is the wage rate, and Π_t is profits from intermediate goods prices.

The growth rate of the money supply depends on the shock s and is given by an increasing function $\mu(s)$.

A. Define a *recursive equilibrium* for this economy. Specifically, this is an equilibrium in which the continuation equilibrium for any history depends only on the current shock s.

SOLUTION: ■

B. To do so, set up the Bellman equation for consumers. What are the state variables for each consumer? What are the state variables for the economy as a whole? Set up the problems for the three types of firms. What are the market clearing conditions?

SOLUTION: ■

C. Consider a version of this economy with no shocks. Characterize the equilibrium.

SOLUTION: ■

D. Now suppose there is a shock only in period 1. Characterize the equilibrium for this economy. (You may assume the equilibrium is as in part 3 from period 1 onwards.)

SOLUTION: ■

E. (Extra Credit.) Characterize the equilibrium with i.i.d. shocks in each period.

SOLUTION: ■

4 ECON 8107 (Amador)

4.1 Hugget Model

4.2 Aiyagari Model

4.3 Exercises

4.3.1 Prelim: Spring 2025 (Incomplete Markets, Labor Supply, and Aggregate Risk)

This question is related to the 2025 8108 final. There is a mass one of households. The household's discount factor is $\beta \in (0, 1)$ and their period utility function is given by

$$u(c, n) = \log(c) - \psi n,$$

where c is consumption, n is labor supply, and $\psi > 0$.

A household supplies zn units of effective labor to the market at a wage w_t and where z is stochastic. It can take one of two values: z_H and z_L, with $z_H > z_L > 0$. The probability of being in state z_H is π_z and the probability of being in state z_L is $1 - \pi_z$. The realization is idiosyncratic, independent across households and time. Households also have access to a one-period risk-free uncontingent asset that pays a gross interest rate R_t. A household faces the following budget constraint:

$$c + a' = R_{t-1}a + w_t zn,$$

where a are assets, and has the following (ad-hoc) borrowing constraint:

$$a' \geq -\phi,$$

where $\phi > 0$. In what follows we allow the household's choice of n to be in $(-\infty, \infty)$.

Firms operate in a competitive environment and have a linear production function to produce the final consumption good:

$$Y = A_t N,$$

where N are the *total effective units of labor* and A_t is aggregate productivity. The aggregate productivity A_t is realized at the beginning of period t, taking value A_H with probability π_A and A_L with probability $1 - \pi_A$, with $A_H > A_L > 0$. This realization is iid.

We will discuss the competitive equilibrium of this incomplete markets economy.

(a) Show that the consumption of a household with state z when productivity is A is:

$$c(z, A) = \frac{zA}{\psi}.$$

SOLUTION: First, we know that since firms operate in a competitve environment, it must be that wages are equal to marginal returns to labor. Specifically, we know that $w = Y_n = A_t$. With this in mind, we can then write the following bellman equation

$$\begin{aligned} V(z, A, a) &= \max_{c,n,a'} \{\log(c) - \psi n + \beta \mathbb{E}[V(z', A', a')]\} \\ s.t. \quad & c + a' = Ra + Azn. \end{aligned}$$

The budget constraint allows us to remove n which let's us rewrite

$$V(z, A, a) = \max_{c,a'} \left\{ \log(c) - \frac{\psi}{Az} (c + a' - Ra) + \beta \mathbb{E}[V(z', A', a')] \right\}$$

Now, let's take a look at the first order condition for c. We get

$$\frac{1}{c} - \frac{\psi}{Az} = 0 \implies c(z, A) = \frac{zA}{\psi}.$$

■

(b) The equilibrium interest rate R_t is stochastic, but only a function of the realized aggregate productivity level A. Find $R(A)$. **Hint:** *Use that households with the lowest z, z_L, are always borrowing constrained.*

SOLUTION: Let's first derive the Euler equation. Starting with the envelope condition with respect to assets and then plugging this into the FOC with respect to a', we can write.

$$
\begin{aligned}
V_a(z',A',a') &= \frac{\psi R}{Az}, \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}[V_a(z',A',a')], \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}\left[\frac{\psi R'}{A'z'}\right], \\
\implies \frac{1}{Az} &= \beta \mathbb{E}\left[\frac{R'}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az \mathbb{E}\left[\frac{1}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right).
\end{aligned}
$$

We have now boiled down R as an expression of z and A (note also that I have conjectured that R is time invariant which I will later confirm). Then, using the hint, we know that households that have been hit with z_L are borrowing constrained and therefore may not necessarily have a binding Euler condition (given that they may want to borrow more). On the other hand, individuals who have been hit with the high productivity shock are guaranteed to have a binding Euler condition (no matter their previous asset holdings) by the asset market clearing condition. As such, we can then continue to write

$$
\begin{aligned}
\frac{1}{R'} &= \beta Az \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right), \\
\implies \frac{1}{R'} &= \beta Az_H \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right), \\
\implies R' &= \frac{1}{\beta Az_H} \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right)^{-1}.
\end{aligned}
$$

Since this expression is independent of time period (which confirms the conjecture), we can write

$$
\boxed{R(A) = \frac{1}{\beta Az_H} \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right)^{-1}.}
$$

■

(c) Let's introduce a government that issues a constant amount of bonds $B > 0$ in every period and uses lump-sum transfers/taxes to balance its budget. What happens to the equilibrium interest rate after this policy?

SOLUTION: The **equilibrium interest rate would stay the same**. To show this, we can look to see what changes and what stays the same.

- Changes - BC and Assets Market Clearing: Now households are faced with two types of savings mehcanisms and also are hit with a transfer/tax. Specifically their budget constraint now becomes (we know that the rate of return for asset holdings and government bonds must be the same by no arbitrage conditions)

$$c + a' + b' = R_{t-1}a + R_{t-1}b + A_t zn + T_t,$$

where the government budget constraint is now instead

$$B = R_{t-1}B + T_t.$$

Next, the asset's market clearing is

$$B = \sum_{z,a} b'(z,a) + a'(z,a).$$

These conditions, however, don't enter the euler condition and therefore will not have an effect on the value of R.

- Same - Euler Condition: We derived this in a manner that was independent of asset holdings. Specifically we know that since both assets a' and b' earn the same return, their handling in the first order conditions and envelope condition are the same and we will still get an expression for R that is solely dependent on A.

■

(d) Under what conditions is such a policy certain to deliver a Pareto improvement to all households? How does your answer here differ from the case with no aggregate risk? Explain your answers.

SOLUTION: We know that $B > 0$ implies that these government bonds are a savings mechanism for households, rather than a borrowing mechanism. Additionally, we know from the government budget constraint that

$$B = R(A)B + T_t \implies T_t = (1 - R(A))B.$$

If $R(A) > 1$ then $T_t < 0$ and if $R(A) < 1$ then $T_t > 0$. This is important since we know that there are some households that are borrowing constraint. A positive transfer would guaranteed lessen the number of borrowing constrained individuals since their in-period income increases whereas a negative transfer would cause more people to become borrowing constrained. More and less people being borrowing constraint is important for welfare because it implies that a given household is not optimally consumed according to their preferences. As such, this government bond policy would certainly deliver a pareto improvement if $\boldsymbol{R(A) < 1}$ for all $A \in \{A_H, A_L\}$.

Since this is an incomplete markets model, risk incentivizes precautionary savings and the interest rate reflects this fact. We know that removing the aggregate risk from this economy would reduce overall risk and decrease the need for precautionary savings, thereby increasing $R(A)$ and making it less likely that $R(A) < 1$ holds. Removing aggregate risk makes it **less likely that the government policy is a pareto improvement**. ■

4.3.2 Prelim: Fall 2025 (Incomplete Markets, Labor Supply, and Aggregate Risk)

This question is based on the 2025 8108 final and the May prelim. There is a mass one of households. The household's discount factor is $\beta \in (0, 1)$ and their period utility function is given by

$$u(c, n) = \log(c) - \psi n$$

where c is consumption, n is labor supply, and $\psi > 0$.

A household supplies zn units of effective labor to the market at a wage w_t and where z is stochastic. It can take one of two values: z_H and z_L, with $z_H > z_L > 0$. The probability of being in state z_H is π_z and the probability of being in state z_L is $1 - \pi_z$. The realization is idiosyncratic, independent across households and time. Households also have access to a one-period risk-free uncontingent asset that pays a gross interest rate R_t. A household faces the following budget constraint:

$$c + a' = R_{t-1}a + w_t z n,$$

where a are assets, and has the following (ad-hoc) borrowing constraint $a' \geq -\phi$, where $\phi > 0$. In what follows we allow the household's choice of n to be in $(-\infty, \infty)$.

Firms operate in a competitive environment and have a linear production function to produce the final consumption good:

$$Y = A_t N$$

where N are the *total effective units of labor* and A_t is aggregate productivity. The aggregate productivity A_t is realized at the beginning of period t, taking value A_H with probability π_A and A_L with probability $1 - \pi_A$, with $A_H > A_L > 0$. This realization is iid.

We will discuss the competitive equilibrium of this incomplete markets economy. In such an equilibrium, the consumption of a household with z units of labor when aggregate productivity is A is $c(z, A) = zA/\psi$.

(a) Why is this an incomplete markets economy?

SOLUTION: This is an incomplete markets economy simply because although there are different potential realizations (multiple states) across aggregate and idiosyncratic utility, the household can only trade bonds that are uncontingent of these realizations. ■

(b) Show that the equilibrium interest rate $R_t = R(A_t)$ where the function R is given by:

$$R(A) = \frac{1}{\beta}\left(\frac{1}{\pi_A \frac{A}{A_H}\left(\pi_z \frac{z_H}{z_H} + (1-\pi_z)\frac{z_H}{z_L}\right) + (1-\pi_A)\frac{A}{A_L}\left(\pi_z \frac{z_H}{z_H} + (1-\pi_z)\frac{z_H}{z_L}\right)}\right).$$

Hint: Use that the household with the lowest z, z_L, is borrowing constrained, and thus the Euler equation of the household with the highest z must hold.

SOLUTION: Let's first derive the Euler equation. Starting with the envelope condition with respect to assets and then plugging this into the FOC with respect to a', we can write.

$$\begin{aligned}
V_a(z', A', a') &= \frac{\psi R}{Az}, \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}[V_a(z', A', a')], \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}\left[\frac{\psi R'}{A'z'}\right], \\
\implies \frac{1}{Az} &= \beta \mathbb{E}\left[\frac{R'}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az \mathbb{E}\left[\frac{1}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az\left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right).
\end{aligned}$$

We have now boiled down R as an expression of z and A (note also that I have conjectured that R is time invariant which I will later confirm). Then, using the hint, we know that households that have been hit with z_L are borrowing constrained and therefore may not necessarily have a binding Euler condition (given that they may want to borrow more). On the other hand, individuals who have been hit with the high productivity shock are guaranteed to have a binding Euler condition (no matter their previous asset holdings) by the asset market clearing condition. As such, we can then continue to write

$$\begin{aligned}
\frac{1}{R'} &= \beta A z \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L} \right), \\
\implies \frac{1}{R'} &= \beta A z_H \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L} \right), \\
\implies R' &= \frac{1}{\beta A z_H} \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L} \right)^{-1}, \\
\implies R' &= \frac{1}{\beta A z_H} \left(\frac{\pi_A}{A_H} \left(\frac{\pi_z}{z_H} + \frac{1-\pi_z}{z_L} \right) + \frac{1-\pi_A}{A_L} \left(\frac{\pi_z}{z_H} + \frac{1-\pi_z}{z_L} \right) \right)^{-1},
\end{aligned}$$

Since this expression is independent of time period (which confirms the conjecture), we can distribute the A and z_H to write

$$\boxed{R(A) = \frac{1}{\beta} \left(\frac{1}{\pi_A \frac{A}{A_H} \left(\pi_z \frac{z_H}{z_H} + (1-\pi_z) \frac{z_H}{z_L} \right) + (1-\pi_A) \frac{A}{A_L} \left(\pi_z \frac{z_H}{z_H} + (1-\pi_z) \frac{z_H}{z_L} \right)} \right).}$$

■

(c) Suppose that $R(A) < 1$ for all $A \in \{A_L, A_H\}$. Suppose that a government issues a constant level of debt B financed by lump sum taxes/transfers. This policy generates a *Pareto improvement* over the initial equilibrium. Explain why.

SOLUTION: We know that the bonds are constant over time and also that, conditioned on A, the interest rate is time invariant. The resulting government budget constriant is

$$B = R(A)B + T \implies (1 - R(A))B = T.$$

Since $R(A) < 1$, we know that $1 - R(A)$ is positive. As such, there are then three cases.

- $\underline{B = 0}$: The trivial case, nothing changes.
- $\underline{B > 0}$: If bonds are positive, then transfers will also be positive. This pulls some people away from the borrowing constraint and is a pareto improvement.
- $\underline{B < 0}$: This is not a pareto improvement for the opposite reasoning as above. As such the policy is only a pareto improvement when $B > 0$.

■

4.3.3 Final: Spring 2025 (Aiyagari economy with a storage technology)

Consider a standard Aiyagari economy. There is a continuum of mass one of households with utility $u(c)$ (strictly increasing, strictly concave) and discount factor $\beta \in (0,1)$. Each household receives an idiosyncratic labor productivity shock $z \in S = \{z_1, z_2, \dots, z_N\}$, with all $z_i > 0$, which follows a Markov chain with transition matrix Π that has full support (all entries strictly positive). Shocks are independent across households and the law of large numbers applies.

A representative firm operates a neoclassical production function $F(K, L)$ with constant returns to scale, where K is capital and $L = \bar{z}$ is aggregate effective labor (constant in the stationary equilibrium). Capital depreciates at rate $\delta \in (0,1)$. Competitive factor prices are:

$$R(K) = F_K(K, L) + 1 - \delta, \qquad w(K) = F_L(K, L).$$

Households can save in capital ($a' \geq 0$) but cannot borrow. The household's budget constraint is:

$$c + a' = Ra + wz, \qquad a' \geq 0,\ c \geq 0.$$

Part I: The standard economy.

Denote the stationary equilibrium of this economy by $(R^\circ, w^\circ, K^\circ)$. $\boxed{\textbf{Assume that } R^\circ < 1.}$

(a) Write down the Bellman equation for a household in the stationary equilibrium. What are the state and choice variables?

SOLUTION: ■

(b) Is the borrowing constraint $a' \geq 0$ an ad-hoc or a natural borrowing limit? Explain.

SOLUTION: ■

(c) Briefly explain why $R^\circ < 1$ is possible in the Aiyagari model but not in the representative-agent neoclassical model.

SOLUTION: ■

(d) Use the Aiyagari diagram (assets A on the x-axis, gross interest rate R on the y-axis) to illustrate the stationary equilibrium with $R^\circ < 1$. Label the asset supply curve $A(R)$ and the capital demand curve $K(R)$.

SOLUTION: ■

Part II: Introducing storage.

Suppose households now gain access to a **storage technology**: one unit of the consumption good stored today yields one unit tomorrow. That is, storage pays a gross return of 1. Storage is available in perfectly elastic supply (households can store any non-negative amount).

Let k denote a household's holdings of capital and s its holdings of storage. The budget constraint becomes:

$$c + k' + s' = Rk + s + wz,$$

with $k' \geq 0$, $s' \geq 0$, and $c \geq 0$. Neither asset can be held in negative amounts.

(e) Write down the Bellman equation for a household with access to both capital and storage.

SOLUTION: ■

(f) Define a stationary equilibrium for this economy. Be precise about the market clearing conditions.

SOLUTION: ■

(g) Show that when $R > 1$, no household holds storage in equilibrium (i.e., $s' = 0$ for all). When $R = 1$, argue that households are indifferent between capital and storage, so only the *total* saving $a' = k' + s'$ matters.

SOLUTION: ■

(h) Argue that in any equilibrium, $R \geq 1$. What would happen if $R < 1$?

SOLUTION: ■

(i) Let $A(R, w)$ denote the total asset supply (capital plus storage) as a function of the gross return R and wage w. In the standard Aiyagari diagram, w is determined by R through the factor price frontier: given R, capital K is determined by $R = F_K(K, L) + 1 - \delta$, and the wage is $w(R) = F_L(K, L)$. Write $A(R) \equiv A(R, w(R))$.

Show how the storage technology modifies the capital supply curve in this diagram. In particular, explain why the modified curve has the following shape (with K on the horizontal axis and R on the vertical):

(i) a *flat* segment at $R = 1$ running from $K = 0$ to $K = A(1)$, and

(ii) the standard upward-sloping $A(R)$ for $R > 1$ (to the right of $K = A(1)$).

Draw the modified diagram, add the capital demand curve $K(R)$, and identify the new equilibrium. Where on the supply curve does the equilibrium occur given $R^\circ < 1$?

SOLUTION: ■

(j) How does the introduction of storage affect the equilibrium capital stock and wages compared to the original economy (where $R^\circ < 1$)? What happens to the excess savings that would have driven R below 1?

SOLUTION: ■

(k) How does the new steady-state capital stock compare to the golden rule level?

Hint: The golden rule capital stock K_{gold} is defined by $F_K(K_{\text{gold}}, L) = \delta$.

SOLUTION: ■

Part III: Welfare and policy.

We now consider the welfare and policy implications of the arrival of the storage technology.

(l) Would introducing the storage technology generate a Pareto improvement relative to the original $R^\circ < 1$ equilibrium? Discuss. *Hint:* Think of the wage.

SOLUTION: ■

(m) Consider a policy where the government, upon the arrival of the storage technology:

(i) taxes both capital returns and storage returns so the household faces a common after-tax gross return $R' = R^\circ$,

(ii) lets capital fall to a new level $K' < K^\circ$,

(iii) uses a wage subsidy to maintain the effective wage at w°.

Argue that the government's net revenues from the three instruments take the form:

$$\begin{aligned}
\text{Capital tax revenue} &= \left[R(K_t) - R^\circ\right] K_t, \\
\text{Storage tax revenue} &= \left(1 - R^\circ\right) S_t, \\
\text{Wage subsidy cost} &= \left[w^\circ - w(K_t)\right] L,
\end{aligned}$$

where K_t is the capital stock and S_t is aggregate storage at date t.

Using the firm's first-order conditions, compute each of these three quantities separately for $t = 0$ and for $t > 0$ (the new steady state). For $t > 0$, first argue how much storage S the economy holds.

Note: Because capital is a **predetermined variable**, treat $t = 0$ and $t > 0$ separately. At $t = 0$ the capital stock is still K°; only from $t = 1$ onward does capital equal K'.

Hint: At $t = 0$: what is K_0? How much storage S_0 is there? For $t > 0$: why are total household savings still K°, and what does that imply for $S = K^\circ - K'$?

SOLUTION: ■

(n) **[Diamond-style policy: fiscal surplus.]**

Using the revenues and costs from (m), write the government's net fiscal surplus at $t > 0$. Show that it simplifies to

$$\text{Surplus} = \left[F(K', L) - \delta K'\right] - \left[F(K^\circ, L) - \delta K^\circ\right],$$

and argue that this is *positive* for any $K' \in [K_{\text{gold}}, K^\circ)$.

Hint: Use the CRS identity $F(K, L) = F_K(K, L)\, K + F_L(K, L)\, L$ to simplify.

Explain why this policy can lead to a Pareto improvement.

SOLUTION: ■

5 ECON 8108 (Luttmer)

I've never done any continuous time macro before this class and so getting intuition about what is going on is taking me some time. These notes will likely be very detailed as I'm writing them in order to learn myself.

5.1 The canonical continuous time consumer problem

Here, **preferences** are

$$\int_0^\infty e^{-\rho t} U(C_t)$$

for some well behaving $U(C_t)$. Off the get go, the main thing to note that instead of the β temporal discounting that we had before, the continuous time equivalent is $e^{-\rho t}$ where we should note that $\int_0^\infty e^{-\rho t}\, dt = \frac{1}{\rho}$ for $\rho > 0$. Next, the present-value **budget constraint** is (think an Arrow-Debreu world) is

$$\int_0^\infty \pi_t C_t\, dt \leq \pi_0 W_0.$$

We are abstracting right now from firms or income processes. We are looking at lifetime wealth W_0 and prices π_t as some exogenous feature. So with this in mind, let's write down the lagrangian

$$\mathcal{L}(C_t, \lambda) = \int_0^\infty e^{-\rho t} U(C_t)\, dt - \lambda \left(\int_0^\infty \pi_t C_t\, dt - \pi_0 W_0 \right).$$

Which leads to the following two first order conditions

$$\begin{aligned}[C_t]: &\quad e^{-\rho t} DU(C_t) = \lambda \pi_t \\ [\lambda]: &\quad \pi_0 W_0 = \int_0^\infty \pi_t C_t \, dt.\end{aligned}$$

Taking the first FOC, we can write

$$\begin{aligned} e^{-\rho t} DU(C_t) &= \lambda \pi_t, \\ DU(C_t) &= e^{\rho t} \lambda \pi_t, \\ C_t &= [DU]^{-1} \left(\lambda \pi_t e^{\rho t} \right). \end{aligned}$$

Putting this into the second first order conditions, we get

$$\pi_0 W_0 = \int_0^\infty \pi_t [DU]^{-1} \left(\lambda \pi_t e^{\rho t} \right) \, dt$$

with which we can solve for $\lambda > 0$. To do a concrete example, take CES utility

$$U(C) = \frac{C^{1-1/\epsilon}}{1-1/\epsilon} \implies DU(C) = C^{-1/\epsilon} \implies [DU]^{-1}(A) = A^{-\epsilon}.$$

Putting this into the expression above, we get that

$$\begin{aligned} \int_0^\infty \pi_t [DU]^{-1} \left(\lambda \pi_t e^{\rho t} \right) \, dt &= \pi_0 W_0 \\ \int_0^\infty \pi_t \left(\lambda \pi_t e^{\rho t} \right)^{-\epsilon} \, dt &= \pi_0 W_0 \\ \int_0^\infty \lambda^{-\epsilon} \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt &= \pi_0 W_0 \\ \lambda^{-\epsilon} \int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt &= \pi_0 W_0 \\ \lambda^{-\epsilon} &= \frac{\pi_0 W_0}{\int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt} \\ \lambda &= \left[\frac{\pi_0 W_0}{\int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt} \right]^{-1/\epsilon} \end{aligned}$$

We can plug in the original expression that we derived for C_t above and we get that

$$\pi_t C_t = \frac{e^{\epsilon \rho t}}{\int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt} \times \pi_0 W_0 \xrightarrow{\epsilon \to 1} \pi_t C_t = \rho^{-\rho t} \pi_0 W_0.$$

This finally implies that $C_0 = \rho W_0$.

All the discussion thus far has been for the present value budget constraint case. We could also, however, consider the same maximization problem but now with a sequential markets budget constraint

$$dW_t = (r_t W_t - C_t)\,dt \iff DW_t = r_t W_t - C_t,$$

where r_t represents some measure of an interest rate. Specifically, we would have

$$\pi_t = \pi_0 \exp\left(-\int_0^t r_s\,ds\right).$$

5.2 Kass-Coopmans

Now, let's introduce some production and investment dynamics into the model. This time, consider some utility function $U(\cdot)$ that is strictly increasing, concave, and sufficiently smooth. This economy contains one representative agent who has **preferences**, for some $\rho > 0$,

$$\int_0^\infty e^{-\rho t} U(C_t)\,dt.$$

We will have production, but we abstract from having a firm that manages production. Specifically, households will have the chance to choose between dedicating their available resources between consumption and capital investment. There is a production function $F(k, l)$ that is strictly increasing, concave, sufficiently smooth, and homogeneous of degree 1. We will also denote $f(k) = F(k, 1)$. Finally let X_t be the amount of investment at time t and δ be the depreciation rate. These definition all help define the **resource contraints** that the consumer faces

$$\begin{aligned} C_t + X_t &\leq f(K_t), \\ DK_t &\leq -\delta K_t + X_t, \\ C_t, K_t &\geq 0. \end{aligned}$$

Combining all of these constraints gives us the one collapsed constraint

$$DK_t \leq f(K_t) - \delta K_t - C_t.$$

Now let's look at the lagrangian now and derive the first order constraints that can help us solve this model.

$$\mathcal{L}(C_t, K_t, \lambda_t) = \int_0^\infty e^{-\rho t} U(C_t)\,dt + \int_0^\infty e^{-\rho t} \lambda_t (f(K_t) - \delta K_t - C_t - DK_t)\,dt.$$

Since we don't like to work with DK_t, let's do integration by parts.

$$\begin{aligned} \int_0^\infty e^{-\rho t} \lambda_t DK_t\,dt &= e^{-\rho T} \lambda_T K_T \Big|_{T=0}^{\infty} - \int_0^\infty e^{-\rho t} K_t (D\lambda_t - \rho \lambda_t)\,dt, \\ &= \lim_{T \to \infty} e^{-\rho T} \lambda_T K_T - \lambda_0 K_0 - \int_0^\infty e^{-\rho t} K_t (D\lambda_t - \rho \lambda_t)\,dt. \end{aligned}$$

Using this, we can then rewrite the lagrangian as

$$
\begin{aligned}
\mathcal{L}(C_t, K_t, \lambda_t) &= \int_0^\infty e^{-\rho t} U(C_t)\, dt + \int_0^\infty e^{-\rho t} \lambda_t (f(K_t) - \delta K_t - C_t - DK_t)\, dt, \\
&= \int_0^\infty e^{-\rho t} \left[U(C_t) + \lambda_t (f(K_t) - \delta K_t - C_t) \right] dt - \int_0^\infty e^{-\rho t} \lambda_t D K_t\, dt, \\
&= \lambda_0 K_0 - \lim_{T\to\infty} e^{-\rho T} \lambda_T K_T + \int_0^\infty e^{-\rho t} \left[U(C_t) + \lambda_t (f(K_t) - (\delta+\rho) K_t - C_t) + K_t D\lambda_t \right] dt, \\
&= \lambda_0 K_0 + \int_0^\infty e^{-\rho t} \left[U(C_t) + \lambda_t (f(K_t) - (\delta+\rho) K_t - C_t) + K_t D\lambda_t \right] dt.
\end{aligned}
$$

Note, we use the transversality condition to claim $\lim_{T\to\infty} e^{-\rho T} \lambda_T K_T = 0$. Now let's take the first order conditions:

$$
\begin{aligned}
&[C_t] : e^{-\rho t}[U'(C_t) - \lambda_t] = 0 \implies DU(C_t) = \lambda_t \implies C_t = [DU]^{-1}(\lambda_t), \\
&[K_t] : e^{-\rho t}[\lambda_t f'(K_t) - (\delta+\rho)\lambda_t + D\lambda_t] = 0 \implies D\lambda_t = \lambda_t(\rho + \delta - Df(K_t)).
\end{aligned}
$$

Right now, this is a system of two equations has three unkowns in C_t, λ_t and K_t. We can, however, collapse this into a system of two equations of two unknowns using the condensed budget constraint. Specifically we plug in the first FOC condition. This gives us the following system of two differential equations

$$
\begin{aligned}
DK_t &= f(K_t) - \delta K_t - [DU]^{-1}(\lambda_t), \\
D\lambda_t &= \lambda_t(\rho + \delta - Df(K_t)).
\end{aligned}
$$

5.2.1 Solving the System

If you have done some Partial Differential Equations before then you know that a solution to the system above is not some simple concept of neat expressions for K_t and λ_t but rather a dynamic object in itself that we often try to represent using things such as phase diagrams.

The idea is that given an initial condition of capital K_0 and λ_0 (which we can recover from the envelope condition $\lambda_0 = DV(K_0)$ an interpret as the shadow price of capital) we trace out the curve of K_t and λ_t depending on the direction of DK_t and $D\lambda_t$ which changes as K_t and λ_t evolves. This traced out curve is the solution and we can use phase diagrams to help build out this curve (1).

A **steady state solution** is, however, something that we can think about and back out easily. We know that a steady state implies that the equilibrium allocations don't change over time. As such, by definition, we can very directly claim that a steady state implies that $DK_t = 0$, $D\lambda_t = 0$ and we get the new system

$$
\begin{aligned}
0 &= f(K) - \delta K - [DU]^{-1}(\lambda), \\
0 &= \lambda(\rho + \delta - Df(K)).
\end{aligned}
$$

Disregarding the edge case of $\lambda = 0$, we can then back out

$$K = [Df]^{-1}(\rho + \delta),\, C = f(K) - \delta K.$$

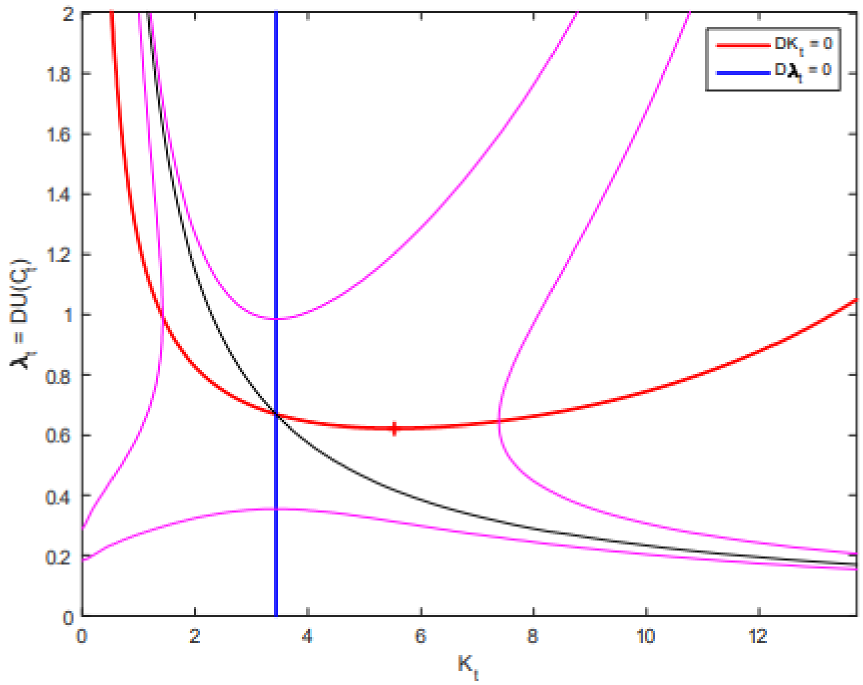

Figure 1: The phase diagram for (K_t, λ_t).

One alternate simple solution class that we can think about is the **golden rule**. That is, choosing a capital stock such that change in capital is offset by depreciation. Specifically, picking some K_G such that

$$Df(K_G) = \delta \iff K_G = [Df]^{-1}(\delta).$$

Since $\rho > 0$ and Df is a decreasing function, we know that the steady state level of capital that we described in the system of equations above must be smaller than the golden rule capital. As such, we know that this steady state level of capital is not optimal.

5.2.2 Irreversible Investment

Before we dive into competitive eqiulibrium, let's consider the same version of the model above but also require that capital investment decisions must be non-negative. Specifically, preferences are the same but the budget constraints now change to

$$\begin{aligned} C_t + X_t &\leq f(K_t), \\ DK_t &\leq -\delta K_t + X_t, \\ C_t, K_t, X_t &\geq 0. \end{aligned}$$

Starting from the beginning, let's write the lagrangian

$$\mathcal{L}(C_t, X_t, \mu_t, \lambda_t) = \int_0^\infty e^{-\rho t} U(C_t)\, dt + \int_0^\infty e^{-\rho t} \lambda_t (f(K_t) - X_t - C_t) + \int_0^\infty e^{-\rho t} \mu_t (X_t - \delta K_t - DK_t).$$

Once again, since we don't like to work with DK_t, let's do integration by parts.

$$\begin{aligned}\int_0^\infty e^{-\rho t} \mu_t DK_t\, dt &= e^{-\rho T} \mu_T K_T \Big|_{T=0}^{\infty} - \int_0^\infty e^{-\rho t} K_t (D\mu_t - \rho \mu_t)\, dt, \\ &= \lim_{T\to\infty} e^{-\rho T} \mu_T K_T - \mu_0 K_0 - \int_0^\infty e^{-\rho t} K_t (D\mu_t - \rho\mu_t)\, dt.\end{aligned}$$

Using this, we can then rewrite the lagrangian as

$$\begin{aligned}\mathcal{L}(C_t, X_t, \mu_t, \lambda_t) = -\lim_{T\to\infty} e^{-\rho T} \mu_T K_T + \mu_0 K_0 + \int_0^\infty e^{-\rho t} \big[U(C_t) + \lambda_t (f(K_t) - X_t - C_t) \\ + \mu_t (X_t - \delta K_t) + K_t (D\mu_t - \rho \mu_t) \big]\, dt.\end{aligned}$$

Again, with the tranversality condition, we can rewrite

$$\mathcal{L}(C_t, X_t, \mu_t, \lambda_t) = \mu_0 K_0 + \int_0^\infty e^{-\rho t} \left[U(C_t) + \lambda_t (f(K_t) - X_t - C_t) + \mu_t (X_t - \delta K_t) + K_t (D\mu_t - \rho\mu_t) \right] dt.$$

Assuming that the non-negativity conditions don't apply and the utility function satisfies the inada conditions (implying that the other constraints bind as well), we can write the following FOCs

$$\begin{aligned}&[C_t] : e^{-\rho t} [U'(C_t) - \lambda] = 0 \implies U'(C_t) = \lambda_t, \\ &[K_t] : e^{-\rho t} [\lambda_t f'(K_t) - \mu_t \delta + D\mu_t - \rho \mu_t] = 0 \implies \lambda_t Df(K_t) = \mu_t (\delta + \rho) - D\mu_t.\end{aligned}$$

Again, we plug in the first FOC into the second FOC and also write down the condensed household budget constraint to get the following system of equations

$$\begin{aligned}D\mu_t &= \mu_t (\delta + \rho) - DU(C_t) Df(K_t), \\ DK_t &= f(k_t) - C_t - \delta K_t.\end{aligned}$$

This along with the transversality condition, an initial value for K_0, and the following conditions classifies the phase diagram (2) and any equilibria:

$$\begin{aligned}DU(C_t) &\geq \mu_t, \\ f(K_t) &\geq C_t, \\ (DU(C_t) - \mu_t)(f(K_t) - C_t) &= 0.\end{aligned}$$

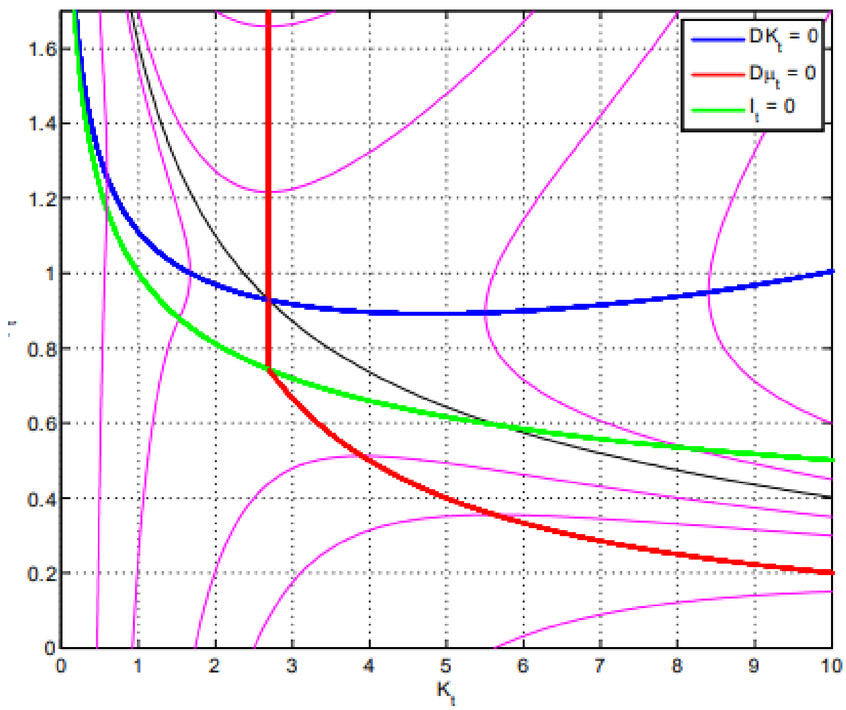

Figure 2: The phase diagram for (K_t, μ_t) with irreversible investment.

5.2.3 Competitive Equilibrium

So far, we have looked at this economy from a social planner's perspective without prices. Let's now try to add firms and prices.

5.2.4 Producers

I will preface, I feel like Erzo's treatment of the intermediate and final good producing firms does not seem very intuitively parallel to the discrete time versions that we have learned about in Chari. The Erzo firm is actually a quite simpler beast with simpler technology, but I think it is the skipping straight to final and intermediate good producers instead of first having them as one block and the reason for doing so that is tripping me up. I'll go really slow.

The **final good producer** takes a homogenous intermediate good g as an input and linearly transforms this into either capital, x, or the final consumption good c. Let q_t be the price of capital, p_t be the price of the intermediate good, and normalize the price of the consumption good to 1. This setup yields the following maximization problem

$$\max_{c,x,g}\{c + q_t x - p_t g\} \quad s.t. \quad g = c + x.$$

Which we can more simply just write as

$$\max_{c,x}\{c + q_t x - p_t(c + x)\}.$$

We know that there can't be unbounded profits, which means that we must have $p_t \geq 1$ and $p_t \geq q_t$ because otherwise we could just buy infinite amount of the intermediate good and produce an infinite amount of either capital or the consumption good. As such, we can clearly state

$$p_t = \max\{1, q_t\}.$$

Now consider, if $q_t > 1$ then $p_t = q_t$ and the final goods producer would have no incentive to produce any consumption good. Since, from the inada conditions, we know that there cannot be an equilibrium with 0 final good consumption, it must be then that an equilibrium has $p_t = 1 \geq q_t$ where there is zero capital production if $q_t < 1$ for the same argument as just stated but visa-versa.

The **intermediate good producer** takes in capital and labor as an input and produces the homogenous intermediate good. Given wages w_t, define

$$v_t = \max_{l}\{F(1, t) - w_t l\}.$$

Since we know $p_t = 1$ and given that capital input here is is non-zero which implies that $q_t = 1$, we can claim that v_t represents the income earned by an intermediate firm who owns one unit of capital. We know that the aggregate labor supply is 1 and the aggregate capital stock is K_t so taking the first order condition for the expression above, we get

$$D_2 F(K_t, 1) = w_t.$$

Then, because F has constant returns to scale, we can claim that $v_t = D_1 F(K_t, 1) = Df(K_t)$ and therefore

$$w_t = f(K_t) - Df(K)K_t.$$

This final good producer may seem quite trivial but the reason of defining one is because of the irreversible investment. Specifically, in these sorts of models with constant returns to scale, the price of capital q_t can be less than 1 unit of the consumption good. This leads to arbitrage scenarios because one can scale production of the consumption good linearly for less than the cost of the consumption good. This would lead to some equilibrium where all resources would be funneled into investment and yield no consumption. There needs to be a mechanism that stops capital prices from being less than one.

5.2.5 Consumers

Preferences are, of course, the same as in planner's problem but the budget constraint does change. Now, the consumer earns labor income w_t, consumes C_t, and can borrow or lend at interest rate r_t for assets A_t. Assets evolve according to

$$DA_t = r_t A_t + w_t - C_t.$$

Erzo doesn't write this derivation out explicitly but I tend to get confused if every step is not spelled out, so we can then rewrite the asset evolution expression as

$$\begin{aligned}
DA_t &= r_t A_t + w_t - C_t, \\
DA_t - r_t A_t &= w_t - C_t, \\
\left(-\int_0^t r_u\, du\right)[DA_t - r_t A_t] &= \left(-\int_0^t r_u\, du\right)[w_t - C_t], \\
\left(-\int_0^t r_u\, du\right) DA_t - \left(-\int_0^t r_u\, du\right) r_t A_t &= \left(-\int_0^t r_u\, du\right)[w_t - C_t], \\
D\left[\exp\left(-\int_0^t r_u\, du\right) A_t\right] &= \left(-\int_0^t r_u\, du\right)[w_t - C_t].
\end{aligned}$$

Integrating this differential equation from t to $T > t$ we then get

$$\begin{aligned}
\int_t^T D\left[\exp\left(-\int_0^s r_u\, du\right) A_s\right] ds &= \int_t^T \exp\left(-\int_0^s r_u\, du\right)[w_s - C_s]\, ds, \\
\exp\left(-\int_0^T r_u\, du\right) A_T - \exp\left(-\int_0^0 r_u\, du\right) A_t &= \int_t^T \exp\left(-\int_0^s r_u\, du\right)[w_s - C_s]\, ds, \\
\int_t^T \exp\left(-\int_0^s r_u\, du\right) C_s\, ds + \exp\left(-\int_0^T r_u\, du\right) A_T &= A_t + \int_t^T \exp\left(-\int_0^s r_u\, du\right) w_s\, ds.
\end{aligned}$$

Then express everything relative to time t consumption, as opposed to time 0 consumption, we get

$$\int_t^T \exp\left(-\int_t^s r_u\, du\right) C_s\, ds + \exp\left(-\int_t^T r_u\, du\right) A_T = A_t + \int_t^T \exp\left(-\int_t^s r_u\, du\right) w_s\, ds.$$

This gives us a map between a consumer's asset level A_t and their consumption choices from t to T, $\{C_s\}_{s\in[t,T]}$, into A_T. We are getting there slowly, but we have an expression to tell us how consumption plans and current asset holsings can give us future asset holdings. We can also use this expression to classify the borrowing constraint, as $T \to \infty$

$$\lim \inf_{T\to\infty} \exp\left(-\int_t^T rs\, ds\right) A_T \geq 0.$$

Define wealth (aka current assets plus the present value of future income flows) as

$$W_t = A_t + \int_t^\infty \exp\left(-\int_t^s r_u\, du\right) w_s\, ds$$

and we can then claim from the long expression above that

$$\int_t^\infty \exp\left(-\int_t^s r_u\, du\right) C_s\, ds \leq W_t.$$

Qualitatively, this is not too crazy, our time t consumption plans are limited by our time t wealth. But... we have gotten our consumer budget constraint now! Here is the consumer's problem so far

$$\begin{aligned} \max_{C_t} \quad & \int_0^\infty e^{-\rho t} U(C_t) \\ s.t. \quad & \int_0^\infty \exp\left(-\int_0^s r_u\, du\right) C_s\, ds \leq W_0. \end{aligned}$$

with the corresponding lagrangian

$$\mathcal{L}(C_t, \lambda_0) = \int_0^\infty e^{-\rho t} U(C_t) + \lambda_0 \left[W_0 - \int_0^\infty \exp\left(-\int_0^s r_u\, du\right) C_s\, ds \right].$$

Finally we can write the following FOC (wrt to consumption

$$e^{-\rho t} DU(C_t) = \lambda_0 \exp\left(-\int_0^t r_s\, ds\right).$$

Differentiating this will give us our euler condition $r_t = \rho + \theta(C_t) DC_t / C_t$. Next, FINALLY! We can solve for consumption

$$C_t = [DU]^{-1} \left(e^{\rho t} \lambda_0 \exp\left(-\int_0^t r_s\, ds\right)\right).$$

Which let's us plug C_t back into the budget constraint

$$\int_0^\infty \exp\left(-\int_0^s r_u\, du\right) C_s\, ds \leq W_0$$

to solve for λ_0 and plug this back in to fully pin down consumption.

5.2.6 Putting everything together - Equilibrium

Let's work on prices. From the consumer's side, we know that the rate of return on assets is r_t and on the firm's side we know a unit of capital generates factor income equal to v_t. Also capital depreciates at a rate δ which implies a value loss of δq_t per unit of time. This gives us the asset pricing equation

$$r_t q_t = v_t + Dq_t - \delta q_t.$$

There are some derivations that I don't quite understand yet but we ultimately get $v_t = r_t + \delta$. I'll come back to this later. Also something else that I will come back to later is the derivation for the expression that will let us compute the trajectory of capital holdings, $\{K_s\}_{s>t}$,

$$A_t + \int_t^{\infty} \exp\left(-\int_t^s r_u \, du\right) w_s \, ds = \int_t^{\infty} \exp\left(-\int_t^s r_u \, du\right) (f(K_s) - X_s) \, ds.$$

Otherwise, combining a lot of the work above, we get

$$r_t = \rho - \frac{D\lambda_t}{\lambda_t}, \quad q_t = \frac{\mu_t}{\lambda_t}, \quad v_t = Df(K_t), \quad w_t = f(K_t) - Df(K_t)K_t.$$

5.3 Hamiltonians

So far, we have done optimization via writing down the lagrangian and then considering the first order conditions. Hamiltonians offer a more direct means of getting to a system of ODEs that give us our solution. It was incredibly groundbreaking for me when I came to the realization of what was going on. It is a set of UNIVERSAL ODEs (as long as the consumer problem can be written in a specific form that I will get to) where to get the final system, all we need to do is look at our formulation of the maximization problem and create this hamiltonian equation that I will get to and voila, we are done. One proof to show that it works needed and then you can go all willy-nilly with it. If you don't get why this is so cool and useful right now (I didn't either until I talked to Claude a lot about why it was cool and useful), then hopefully you will at the end of this.

I'm going to add some context here that I think is woefully missing from Erzo's notes. Let's start with a simple Cass-Koopmans type model that we dealt with in the section above. The social planner solves the problem (given some initial level of capital K_0)

$$\begin{aligned} \max_{C_t, X_t, L_t, K_t} \quad & \int_0^{\infty} e^{-\rho t} u(C_t, L_t) \, dt \\ s.t. \quad (1) \quad & C_t + X_t \leq F(K_t, L_t), \\ (2) \quad & DK_t = X_t - \delta K_t, \\ (3) \quad & C_t \geq 0, \, 0 \leq L_t \leq 1. \end{aligned}$$

Define $U(K, X) = \max_{C,L} \{u(C_t, L_t) : C + X \leq F(K, L)\}$ and note how we can write the equivalent social planner problem

$$\begin{aligned} \max_{K_t, X_t} \quad & \int_0^{\infty} e^{-\rho t} U(K_t, X_t) \, dt \\ s.t. \quad & DK_t = X_t - \delta K_t. \end{aligned}$$

The following work we will do works for any problem that can be written in the form above such that U is concave, increasing in K, and decreasing in X. Define $e^{-\rho t} q_t$ as the shadow price of captial (and the lagrange multiplier for the constraint on the motion of capital). The corresponding lagrangian is then

$$\mathcal{L}(K_t, X_t, q_t) = \int_0^\infty e^{-\rho t}\left[U(K_t, X_t) + q_t(X_t - \delta K_t - DK_t)\right] dt.$$

Once again, let's do integration by parts on DK_t.

$$\begin{aligned}\int_0^\infty e^{-\rho t} q_t DK_t &= e^{-\rho t} q_t K_t \Big|_{t=0}^\infty - \int_0^\infty e^{-\rho t} K_t (Dq_t - \rho q_t)\, dt, \\ &= \lim_{T\to\infty} e^{-\rho T} q_T K_T - q_0 K_0 - \int_0^\infty e^{-\rho t} K_t (Dq_t - \rho q_t)\, dt.\end{aligned}$$

With this, we can rewrite the lagrangian as

$$\mathcal{L}(K_t, X_t, q_t) = \int_0^\infty e^{-\rho t}\left[U(K_t, X_t) + q_t(X_t - \delta K_t) + K_t(Dq_t - \rho q_t)\right] dt + q_0 K_0 - \lim_{T\to\infty} e^{-\rho T} q_T K_T.$$

From here, define the **Hamiltonian**

$$H(K, q) = \max_X \{U(K, X) + q(X - \delta K)\}.$$

Just for some basic intuition, this construction is simply the utility represented as a function of investment decisions X and capital holdings K plus the value (given that q is the shadow price of capital) of the flow of capital (aka the value of the change of capital). Using this, we can reformulate the lagrangian as

$$\mathcal{L}(K_t, q_t) = \int_0^\infty e^{-\rho t}\left[H(K_t, q_t) + K_t(Dq_t - \rho q_t)\right] dt + q_0 K_0 - \lim_{T\to\infty} e^{-\rho T} q_T K_T.$$

This yields the following first order condition

$$[K_t]: \quad H_1(K_t, q_t) + (Dq_t - \rho q_t) = 0 \implies Dq_t = \rho q_t - DH_1(K_t, q_t).$$

Then from the envelope condition for H with respect to q, we get that

$$DH_2(K, q) = X - \delta K = DK_t.$$

Using these two statements, we can get to the following system of differential equations

$$\begin{aligned}Dq_t &= \rho q_t - DH_1(K_t, q_t), \\ DK_t &= DH_2(K_t, q_t).\end{aligned}$$

And we are done! This system defines the solution to the social planner's problem. H will of course change depending on U and u but we will cover some examples in the exercises section to solidify this concretely.

The crazy part is that we were very general with this derivation. Any model where the planning problem can be writen as

$$\max_{K_t, X_t} \quad \int_0^\infty e^{-\rho t} U(K_t, X_t)\, dt$$
$$s.t. \quad DK_t = X_t - \delta K_t.$$

yields the corresponding solution system of differential equations above.

5.3.1 From Hamiltonians to Competitive Equilibrium

There is a lot more than Luttmer talks about in his notes such as the slopes of the resulting differential equations, what more we can milk from the hamiltonians with the Jacobians, and identifying the stable manifold. You can refer to Erzo's notes for that information but I did want to touch up on how to move from the social planners problem that we just spent some time on towards competitive equilibrium as the Hamilonian method makes this easy as well. Critically, the lagrange multiplier q_t is the shadow price of capital (remember back to Kehoe where we studied how the lagrange multipliers of the social planners problem correspond to the prices in competitive equilibrium). Next, the system of ODEs above gives us solutions for K_t and q_t. The rest of the equilibrium prices and allocations will of course depend on the specific problem and what else is going on beyond the household's problem but consider the Cass-Koopman's model. We can recover everything else from just knowing K_t and q_t.

- $\mathbf{X_t}$, we can recover from the hamiltonian FOC as the solution to $DU_2(K_t, X_t) + q_t = 0$.
- $\mathbf{L_t}$ and $\mathbf{C_t}$, since we already know X_t and K_t, we can recover these from the inner maximization problem of $U(K, X) = \max_{C,L} \{u(C_t, L_t) \,:\, C + X \leq F(K, L)\}$.
- $\mathbf{w_t} = f(K_t) - Df(K_t)K_t$, this is from the firm's problem.
- $\mathbf{v_t} = Df(K_t)$, this is also from the firm's problem.
- $\mathbf{r_t} = \frac{v_t - \delta q_t + Dq_t}{q_t}$, this comes from the identities that we derived in the Cass-Koopman's section.

5.4 The Uzawa two-sector economy

Alright now let's consider a type of economy where the factors of production have a bit more nuance to them. Qualitatively, this will be a scenario where we can split our capital and labor resources into capital/labor used to generate more capital and capital/labor used to generate the consumption good. In other words, we can freely transfer production between consumption and capital; there is different technology that produces either of them.

I'll first set up the model and corresponding dynamics from the social planner's perspective, then look at the competitive equilibrium problem (the formulation of what is going on with the firms is what is interesting), and then talk about relevant steady states. To begin the household has the following **preferences**.

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} U(C_t)\, dt,$$

where $\rho > 0$ and $U(\cdot)$ is strictly increasing and concave. They also are given initial capital $K_0 > 0$ and a fixed supply of labor over time $L > 0$. Here is the **technology** for producing consumption and capital paths,

$$\begin{aligned} K_{c,t} + K_{k,t} &= K_t, \\ L_{c,t} + L_{k,t} &= L_t, \\ C_t &\leq F(K_{c,t}, L_{c,t}), \\ X_t &\leq G(K_{k,t}, L_{k,t}), \\ DK_t &= -\delta K_t + X_t. \end{aligned}$$

Where both G and F exhibit constant returns to scale and all $K_{c,t}, K_{k,t}, L_{c,t}, L_{k,t}$ are non-negative. A normal one sector economy would have $F = G$ but differentiating the production functions creates these two consumption and capital sectors. There are some more assumptions placed on F and G and their relationship but you can refer to the notes for that, I'll try to stick to the intuition and deriving the intuition.

5.4.1 The competitive equilibrium environment

There are prices: v_t factor price of capital services, w_t factor price of labor services, p_t output price of consumption, q_t output price of capital, and r_t discount bond prices. On the production side, since there are constant returns to scale and it is competitive markets, the firms choose capital and labor to minimize costs (and also make 0 profits). Their profit functions looks like

$$\begin{aligned} p_t F(k, l) - v_t k - w_t l \leq 0, \\ p_t G(k, l) - v_t k - w_t l \leq 0. \end{aligned}$$

Which lead to the following set of firm problems

$$\begin{aligned} \min_{k,l} \{v_t k + w_t l \;:\; F(k, l) \geq 1\} \geq p_t, \\ \min_{k,l} \{v_t k + w_t l \;:\; G(k, l) \geq 1\} \geq q_t \end{aligned}$$

If we write $k_{c,t}$ and $k_{k,t}$as the cost-minimizing capital-labor ratios, the allocation of labor can be determined by

$$\begin{bmatrix} k_{k,t} & k_{c,t} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} L_{k,t} \\ L_{c,t} \end{bmatrix} = \begin{bmatrix} K \\ L \end{bmatrix}.$$

Consumption and investment are then

$$C_t = F(k_{c,t}, 1)L_{c,t} > 0, \quad I_t = G(k_{k,t}, 1)L_{k,t} \geq 0.$$

Once again, the price of capital satisfies the asset pricing equations

$$r_t q_t = v_t - \delta q_t + Dq_t,$$

which jointly with the Euler condition implies

$$D\mu_t = (\rho + \delta)\mu_t - \lambda_t v_t.$$

As always, the capital stock evolves according to

$$DK_t = -\delta K_t + G(k_{k,t}, 1)L_{k,t}.$$

Together they jointly describe how the state (K_t, μ_t) evolves.

5.4.2 Writing down the hamiltonian

The corresponding hamiltonian for this problem is

$$H(K, \mu) = \max_{C,I} \{U(C) + \mu(I - \delta K)\},$$

where C and I must satisfy the constraints $C \leq F(K_c, L_c)$, $I \leq G(K_k, L_k)$, $K_c + K_k \leq K$, and $L_c + L_k \leq L$. Critically, this formulation abstracts a little from basing the hamiltonian on specific values of K_k, K_c, L_k, L_c but more so possible values of K and L that can be achieved by any combination of the sectoral specific counterparts. Since we know that $U(\cdot)$ satisfies the inada conditions, we can equivalently write

$$H(K, \mu) = \max_{K_c, L_c, I} \{U(F(K_c, L_c)) + \mu(I - \delta K)\}.$$

We know then that the system of differential equations that dictate the solution to this model are

$$\begin{aligned} D\mu_t &= \rho\mu_t - DH_1(K_t, \mu_t) = \rho\mu_t + \delta\mu_t - DU(C_t)D_1F(K_{c,t}, L_{c,t})), \\ DK_t &= DH_2(K_t, \mu_t) = I_t - \delta K_t. \end{aligned}$$

Note the parallels between backing these back from the hamiltonian versus the the asset pricing above (it is just a matter of substituting $G(\cdot)$ and $F(\cdot)$ into the hamiltonian).

The final thing that I wanted to talk about is backing out the sectoral capital and labor allocations. In steady state $DK_t = 0$ and $D\mu_t = 0$. Using equality of the constraints and the system of equations above, we can write that this economy has a unique steady state, characterized by steady-state capital-labor ratios, k_k^* and k_c^* that are defined by,

$$\delta + \rho = D_1 G(k_x, 1), \quad \frac{D_1 G(k_x, 1)}{D_2 G(k_x, 1)} = \frac{D_1 F(k_c, 1)}{D_2 F(k_c, 1)}.$$

For the golden rule, the conditions are slightly different

$$\delta = D_1 G(k_x, 1), \quad \frac{D_1 G(k_x, 1)}{D_2 G(k_x, 1)} = \frac{D_1 F(k_c, 1)}{D_2 F(k_c, 1)}.$$

5.5 Capital Accumulation and Adjustment Costs

To motivate why we would consider the models I will talk about soon, consider the general growth model in discrete time or the Cass-Koopmans that we have talked about in this Luttmer section. The model is constructed in a such a manner that investment today converts into capital tomorrow one-to-one. There are no frictions or delays in capital formulation. Reality is a bit different, simply investing in the construction of a building today does not mean that the building is added into our capital stock overnight. There are additional costs and things take time.

The idea in this section is to model that by introducing an additional cost to investment flow. Specifically, we introduce function $A(K_t, L_t)$ that affects the resource constraint in the following manner by replacing $C_t + I_t \leq f(K_t, L_t)$ with

$$C_t + A(K_t, I_t) \leq F(K_t, L),$$

where

$$A(K_t, I_t) = a\left(\frac{I_t}{K_t}\right) L_t.$$

Here $a(\cdot)$ is convex, which means that there are increasing returns to scale in I_t/K_t, or in other words, the resource cost increases at I_t increases but get smaller as K_t is large: capital heavy firms require fewer resource costs to raise investment. To pin everything down with a specific setup, let's consider the following model. Households have **preferences**

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} U(C_t)\, dt.$$

Labor is taken to be **inelastic** and equal to L. The **constraints for the social planner's** problem are

$$C_t + A(K_t, I_t) \leq F(K_t, L),$$
$$DK_t \leq I_t - \delta K_t.$$

As mentioned before $A(K_t, I_t) = a\left(I_t/K_t\right) K_t$ where $a(0) = 0$, $a(\cdot)$ is increasing and continuous, and $a(\cdot)$ is frequently taken to be quadratic. We can also see how we can recover the Cass-Koopman model if we let $a(\cdot)$ be linear (or better yet the identity. This is not the only way of modeling adjustment costs but is pretty popular. Alternatively, we could also equivalentely write the same formulation with the following constraints

$$\begin{aligned} C_t + X_t &\leq F(K_t, L), \\ DK_t &\leq G(K_t, X_t) - \delta K_t, \end{aligned}$$

where $G(\cdot)$ is defined as $A(1, G(1, x)) = x$. That being said, let's look to solve the problem from the planners perspective. We can start with writing down the **hamiltonian**.

$$\mathcal{H}(K, \mu) = \max_{C, I \geq 0} \{U(C) + \mu(I - \delta K) \, : \, C \leq F(K, L) - A(K, I)\}.$$

The corresponding **first order conditions** for this hamiltonian (given that λ is the lagrange multiplier for the resource constraint, also assuming that investment is non-zero) are

$$\begin{aligned} [C] &: DU(C) - \lambda = 0, \\ [I] &: \mu - \lambda D_2 A(K, I) = 0 \end{aligned}$$

The corresponding **envelope conditions** are

$$\begin{aligned} D_1 \mathcal{H}(K, \mu) &= -\mu\delta + \lambda(D_1 F(K, L) - D_1 A(K, I)), \\ D_2 \mathcal{H}(K, \mu) &= I - \delta K. \end{aligned}$$

Combining statements all together, we can write out the system of ODEs that classify the solution to this economy

$$\begin{aligned} D\mu_t &= \rho\mu_t - D_1 \mathcal{H}(K, \mu) = \mu_t(\rho + \delta) - DU(C_t)(D_1 F(K_t, L_t) - D_1 A(K_t, I_t)), \\ DK_t &= D_2 \mathcal{H}(K_t, \mu_t) = I_t - \delta K_t. \end{aligned}$$

We can then also pin down consumption and investment from the resource constraint and FOC,

$$\begin{aligned} C_t + A(K_t, I_t) &= F(K_t, L), \\ \mu_t &= DU(C_t) D_2 A(K_t, I_t). \end{aligned}$$

Now, moving onto competitive equilibrium

5.5.1 Using Labor to Produce New Capital

5.6 Vintage Capital

5.7 Unemployment, Search, and Matching

Consider a model with

5.8 Perpetual Youth

This model has a gounding in demographics where the key idea is trying to capture the birth and death of consumers. Specifically, there is a unit measure of consumers, new consumers are born at a rate of θ and also consumers also die randomly at a rate of θ (hence the name perpetual youth because one's mortality is independent of age). One way to interpret this is an OLG with some different structure placed on when consumers are born and die.

Let's consider the discrete case, compute some probabilities, and then take some limits to identify the continuous case parallel. Let Δ be a time interval, and let the probability that a consumer survives one time period be $1 - \theta\Delta$. We know then that the probability that a consumer survives to age (or equivalently, the fraction of a cohort that survives to age) $n\Delta$ is

$$PR[N > n] = (1 - \theta\Delta)^n.$$

Now let $a = n\Delta$ represent the total amount of time and we can write $a/\Delta = n$,

$$PR[N > a/\Delta] = (1 - \theta\Delta)^{a/\Delta}.$$

Let's shoot Δ (aka the duration of the time period) down to 0 to get the continuous time equivalents of what we have done so far

$$\begin{aligned}
\lim_{\Delta \to 0} \left(1 - \theta \frac{a}{a/\Delta}\right)^{a/\Delta} &= \lim_{\Delta \to 0} \left(1 - \theta \frac{a}{a/\Delta}\right)^{a/\Delta}, \\
&= \lim x \to \infty \left(1 - \frac{\theta a}{x}\right)^{x}, \\
&= e^{-\theta a}.
\end{aligned}$$

This also gives us that the average life span in a stationary population is $\frac{1}{\theta}$ and distribution of life spans being exponential with density $\theta e^{-\theta a}$. Now let's get to the economics. Consumer preferences are as follows

$$\mathcal{U}(C) = \int_0^\infty \theta e^{-\theta t} \left(\int_0^t e^{-\rho a} U(C_a)\, da \right) dt.$$

We can interpret this as the expected utility.

5.9 Fiscal Theory of the Price Level

5.10 Exercises

5.10.1 Durable Consumption

Consider an economy in which K_t represents the stock of durable consumption goods at time t, and C_t measures the flow of consumption services generated by the stock of durable consumption goods. Preferences are

$$\int_0^\infty e^{-\rho t}\left(\ln(C_t) - J(L_t)\right) dt.$$

The technology is

$$C_t \leq AK_t$$
$$DK_t = -\delta K_t + zL_t$$

Here, ρ, δ, A, and z are all positive parameters. The function $J : \mathbb{R}_+ \to \mathbb{R}$ is smooth, strictly increasing, and strictly convex.

Consider the competitive equilibrium of this economy. Write w_t for the wage, and q_t for the price of a unit of the durable consumption good, both measured in units of consumption services at time t. The real interest rate (in terms of consumption services) is r_t. A consumer with wealth W_t can consume subject to the sequence of constraints $dW_t = (r_t W_t - C_t)dt$ together with $W_t \geq 0$ at all times.

(a) Explain why aggregate consumption services and the real interest rate must satisfy $r_t = \rho + DC_t/C_t$.

Solution: ■

(b) There will be an asset pricing equation of the form $r_t q_t = v_t + Dq_t - \delta q_t$. What is v_t? What will be the wage w_t?

Solution: ■

(c) Give the differential equation for K_t and q_t/C_t that must hold in any equilibrium. Construct the phase diagram for this differential equation.

Solution: ■

(d) Suppose the economy is in its steady state and there is an unforeseen permanent increase in A at $t = 0$. Describe what happens to $\{K_t, q_t/C_t, C_t, L_t, w_t, q_t, r_t\}_{t\geq 0}$.

Solution: ■

(e) Suppose the economy is in its steady state and there is an unforeseen permanent increase in z at $t = 0$. Describe what happens to $\{K_t, q_t/C_t, C_t, L_t, w_t, q_t, r_t\}_{t\geq 0}$. **Note:** If you want to compute an example to check your logic, use $J(L) = L^{1+1/\psi}/(1+1/\psi)$, for some $\psi > 0$.

Solution: ■

5.10.2 A Simple Consumption-Savings Problem

Define

$$U(W) = \max_{C_t \geq 0} \left(\int_0^\infty \rho e^{-\rho t} C_t^{1-1/\varepsilon} \, dt \right)^{1/(1-1/\varepsilon)}$$

subject to

$$DW_t = rW_t - C_t$$

and $W_t \geq 0$ for all $t \geq 0$, starting from $W_0 = W$ for some given $W > 0$.

(a) Find a candidate solution for $\{C_t\}_{t \geq 0}$ and verify that it is well defined if

$$\rho - \left(1 - \frac{1}{\varepsilon}\right) r > 0$$

Compute $U(W)$.

Solution: First, let $\sigma = 1 - \frac{1}{\epsilon}$ and note that $0 < \sigma < 1$ which implies that $f(x) = x^{1/\sigma}$ is a monotone transformation of x which means we can solve the maximization problem using

$$U(W) = \max_{C_t \geq 0} \int_0^\infty \rho e^{-\rho t} C_t^\sigma \, dt.$$

With this, we can write out the lagrangian

$$\begin{aligned}
\mathcal{L}(C_t, \lambda_t) &= \int_0^\infty \rho e^{-\rho t} C_t^\sigma \, dt + \int_0^\infty \lambda_t e^{-\rho t} \left(rW_t - C_t - DW_t \right) \, dt, \\
&= \int_0^\infty \rho e^{-\rho t} C_t^\sigma + \lambda_t e^{-\rho t} \left(rW_t - C_t - DW_t \right) \, dt, \\
&= \int_0^\infty e^{-\rho t} \left(\rho C_t^\sigma + \lambda_t \left(rW_t - C_t - DW_t \right) \right) \, dt, \\
&= \int_0^\infty e^{-\rho t} \left(\rho C_t^\sigma + \lambda_t \left(rW_t - C_t \right) \right) \, dt - \int_0^\infty \lambda_t e^{-\rho t} DW_t \, dt.
\end{aligned}$$

In order to get rid of the DW_t term, we can do integration by parts

$$\begin{aligned}
\int_0^\infty \lambda_t e^{-\rho t} DW_t \, dt &= \lambda_t e^{-\rho t} W_t \Big|_{t=0}^\infty - \int_0^\infty W_t (D\lambda_t e^{-\rho t} - \rho \lambda_t e^{-\rho t}) \, dt \\
&= \lambda_t e^{-\rho t} W_t \Big|_{t=0}^\infty - \int_0^\infty e^{-\rho t} W_t (D\lambda_t - \rho \lambda_t) \, dt \\
&= -\lambda_0 W_0 + \lim_{t \to \infty} \lambda_t e^{-\rho t} W_t - \int_0^\infty e^{-\rho t} W_t (D\lambda_t - \rho \lambda_t) \, dt \\
&=
\end{aligned}$$

■

Now suppose labor income is $zL > 0$ and assets on hand evolve subject to

$$DA_t = rA_t + zL - C_t$$

and $A_t \geq 0$ for all $t \geq 0$, starting from $A_0 = A$ for some given $A > 0$.

(b) Suppose that $r \geq \rho > (1 - 1/\varepsilon)r$. What is the optimal consumption trajectory?

SOLUTION: ■

(c) Suppose that $r < \rho$. Use the Lagrangian for this problem. Conjecture that there will be some optimal time $T \in (0, \infty)$ such that $A_t = 0$ if and only if $t \in [T, \infty)$. Find the equation that defines T and show that it must have a solution in $(0, \infty)$.

SOLUTION: ■

5.10.3 Prelim: Spring 2024 (Firm Size Distributions)

Consider an economy with a representative household whose preferences over trajectories $\{C_t\}_{t \geq 0}$ are given by $\int_0^\infty e^{-\rho t} U(C_t)\, dt$, where $\rho > 0$ and $U(\cdot)$ is a nice utility function. This household inelastically supplies a flow $\mathcal{L} > 0$ of labor and produces a flow $\mathcal{E} > 0$ of capital from scratch. Starting from some $K_0 > 0$, the resource constraints are

$$C_t = F(K_t, L_t), \qquad L_t + M_t \leq \mathcal{L}, \qquad DK_t = -\delta K_t + G(K_t, M_t) + \mathcal{E},$$

where $\delta > 0$, and where F and G are both constant returns to scale production functions that are increasing and concave, with smooth marginal products that range throughout $(0, \infty)$. Consider competitive equilibria. Write r_t for the risk-free rate, w_t for the wage, and q_t for the price of capital.

(a) Give the conditions for a competitive equilibrium. Briefly explain.

SOLUTION: Since these are complete markets and all agents have perfect information, we know that the competitive equilibrium allocation is equivalent to solving the planner's problem and we can back out the prices later.

$$\begin{aligned}
\max_{C_t, K_t, L_t} \quad & \int_0^\infty e^{-\rho t} U(C_t)\, dt \\
s.t. \quad & (1) \quad C_t = F(K_t, L_t), \\
& (2) \quad DK_t = -\delta K_t + G(K_t, \mathcal{L} - L_t) + \mathcal{E}, \\
& (3) \quad 0 \leq L_t \leq \mathcal{L},\ C_t \geq 0
\end{aligned}$$

■

(b) Show that in a steady state, the steady state capital stock $K_t = K$ must satisfy

$$K = \frac{\mathcal{E}}{\delta - G(1,\, m[q])}, \tag{4}$$

$$K = \frac{\mathcal{L}}{l[q] + m[q]}, \tag{5}$$

where $l[\cdot]$ and $m[\cdot]$ are functions of q that are determined by solving (for l and m) the conditions

$$q = \frac{D_1 F(1,l)}{\rho + \delta - D_1 G(1,m)}, \qquad q = \frac{D_2 F(1,l)}{D_2 G(1,m)}. \tag{6}$$

Solution: ■

(c) Eliminate q from (6) and use this to show that l and m must co-move in the region where $G(1,m) < \delta$. You may want to write $g(m) \equiv G(1,m)$ to simplify the algebra. Use this to show that (4) is increasing in q and (5) is decreasing in q.

Solution: ■

Now suppose the flow $\mathcal{E} > 0$ is an average of single units of "startup" capital that arrive at Poisson rates. All the capital that is produced, directly or indirectly, from one particular unit of startup capital is defined to be a firm. Assume that $\delta = \delta_F + \delta_K$, where $\delta_K > 0$ is the rate at which capital inside a firm depreciates continuously, and $\delta_F > 0$ is the rate at which a firm with all its capital is randomly destroyed.

(d) Determine the steady state measure of firms. Determine the stationary distribution of capital across firms when the equilibrium is such that firms grow at a positive rate. How does a reduction in $\mathcal{E} > 0$ affect the mean of this distribution?

Solution: ■